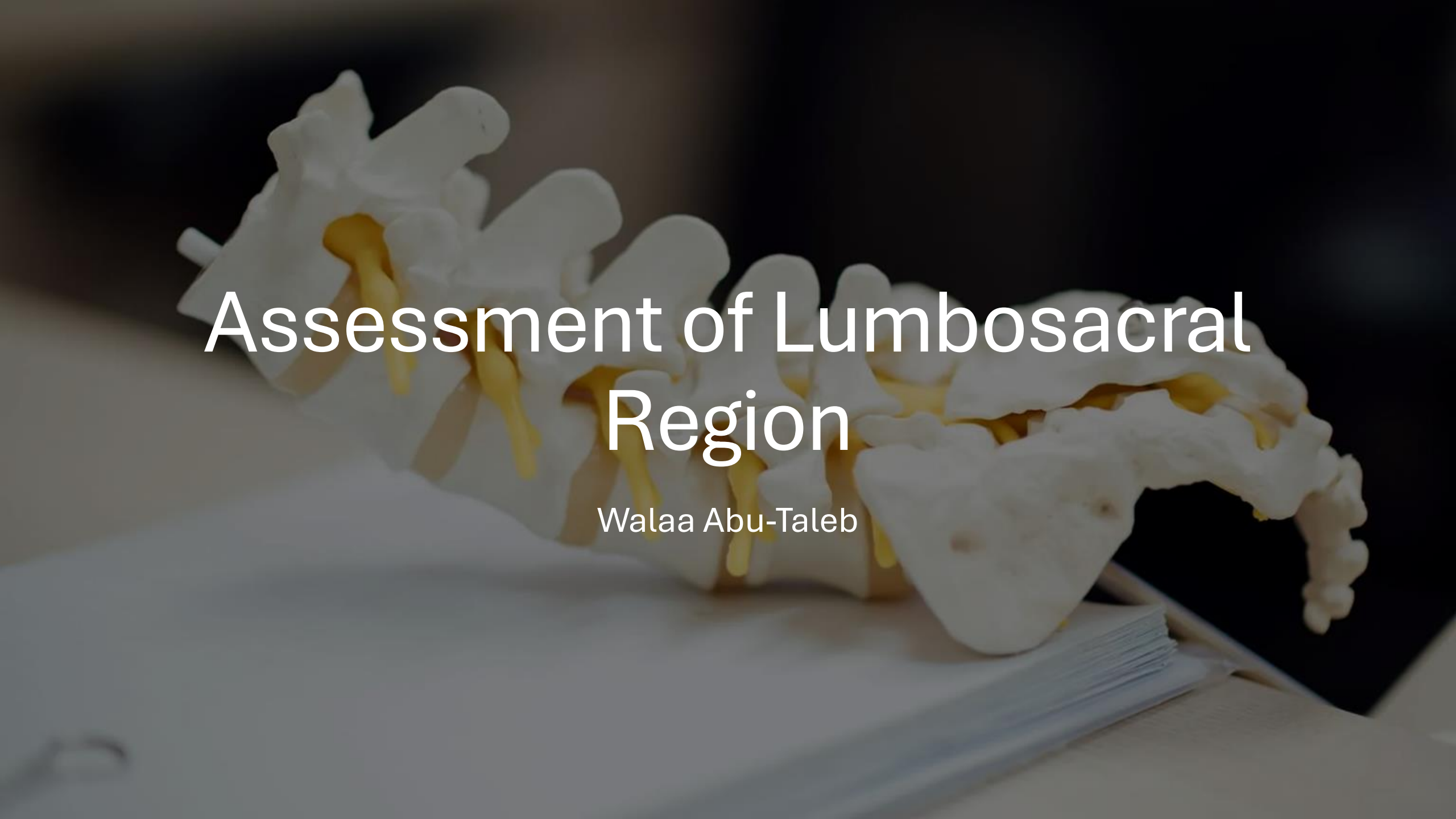
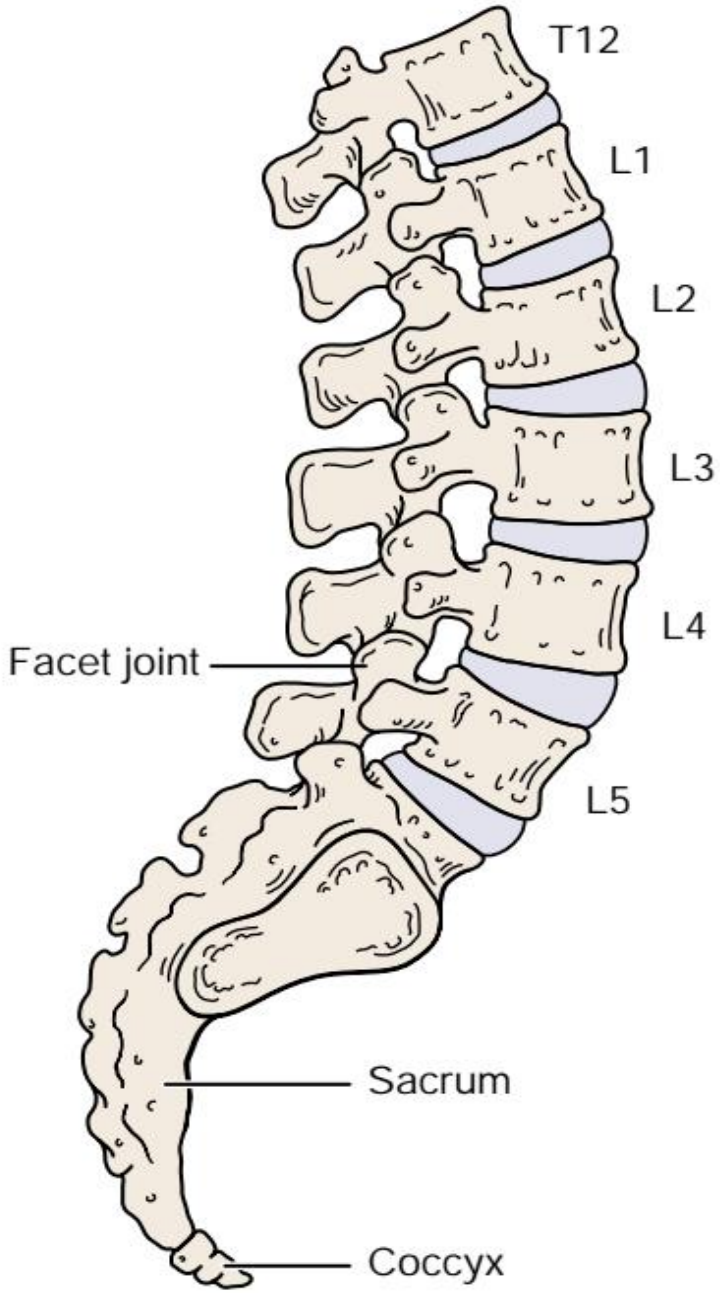


Assessment of Lumbosacral Region

Walaa Abu-Taleb

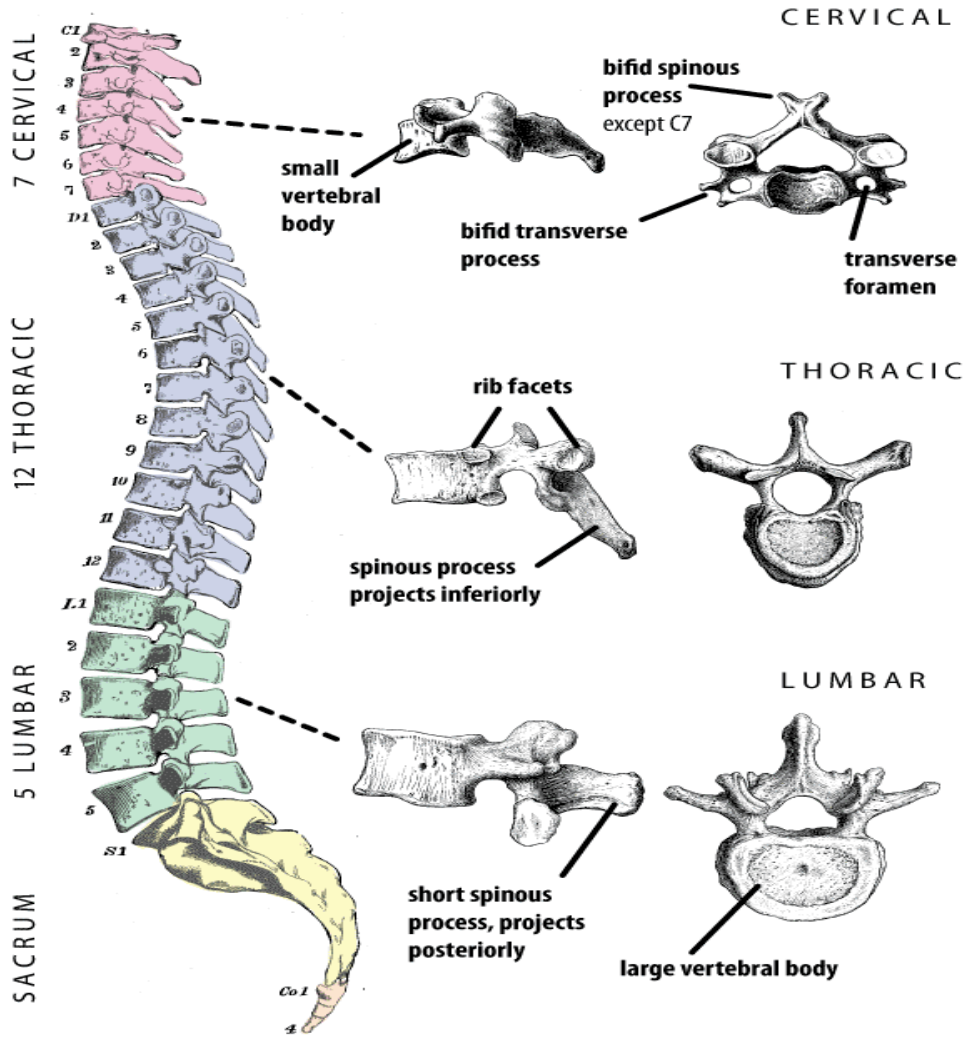
Learning Outcomes

To practice the steps of Objective assessment of the Lumbosacral region.



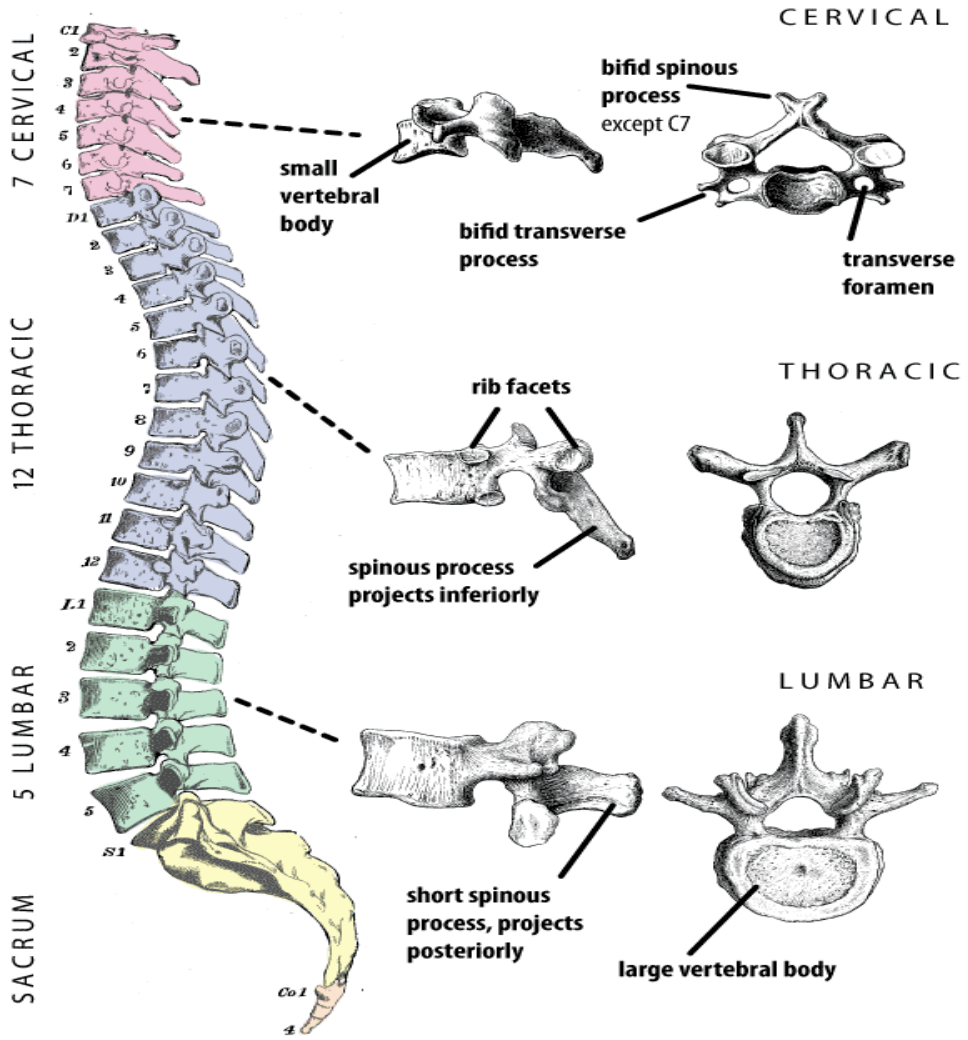
Morphological differences of the vertebrae

Region of Vertebrae	Key Morphological Features	Functional Significance
Cervical (C1–C7)	Small, oval vertebral bodies; large triangular vertebral foramen; transverse foramina in transverse processes; spinous processes short and usually bifid (C2–C6). C1 has no body or spinous process, only arches and lateral masses; C2 has the dens (odontoid process).	Designed for mobility and support of the head. Allows flexion, extension, lateral flexion and rotation while protecting the spinal cord and vertebral arteries.
Thoracic (T1–T12)	Medium-sized, heart-shaped vertebral bodies; circular vertebral foramen; costal facets on bodies and transverse processes for rib articulation; long, inferiorly angled spinous processes.	Provides stability and protection for thoracic organs while allowing limited movement. Adapted for rib attachment and restricted extension.



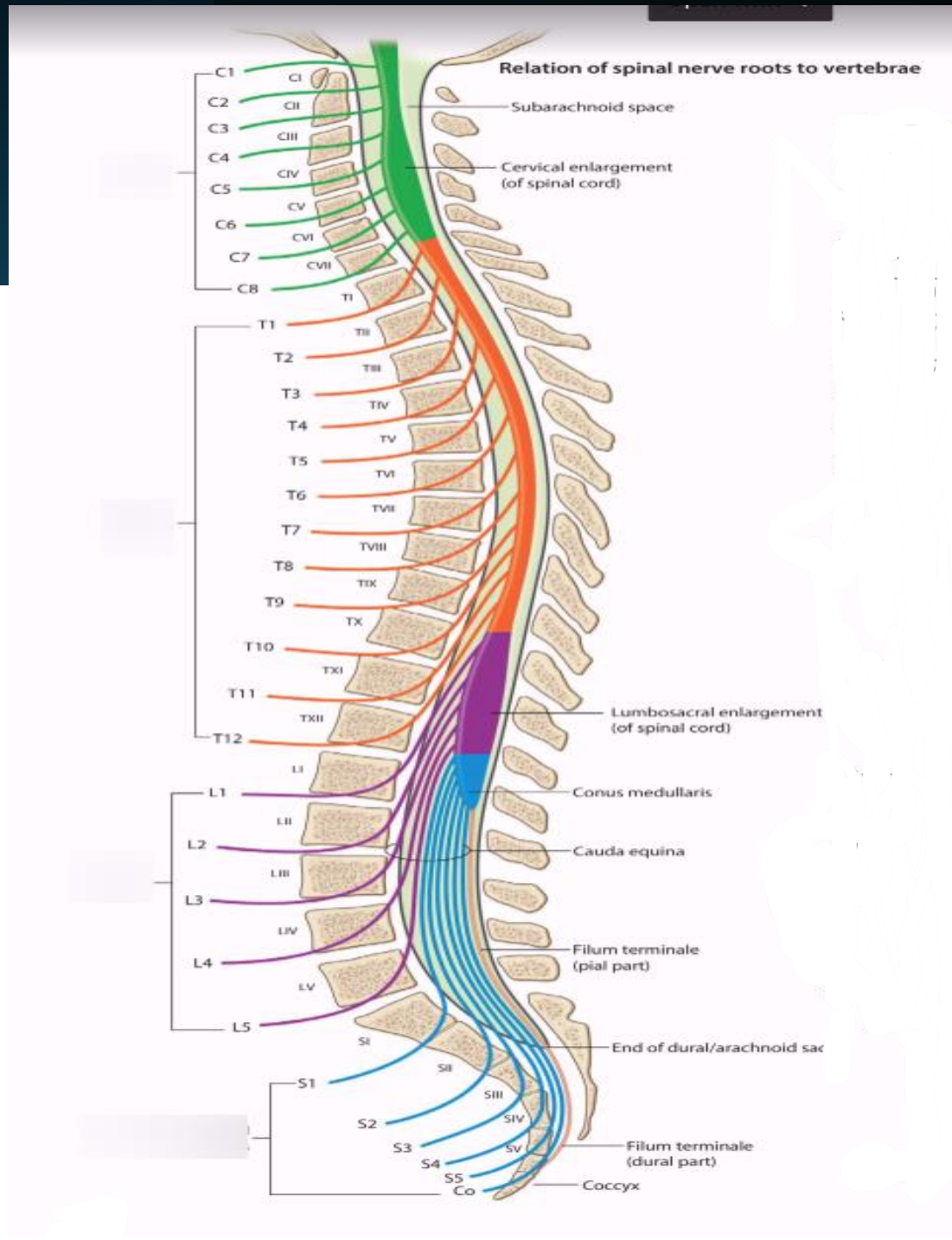
Morphological differences of the vertebrae

Region of Vertebrae	Key Morphological Features	Functional Significance
Lumbar (L1–L5)	Large, kidney-shaped vertebral bodies; triangular vertebral foramen; thick, quadrangular spinous processes; long transverse processes; articular facets oriented mainly in the sagittal plane.	Built for weight-bearing and stability. Allows flexion and extension while limiting rotation to protect the lower spine under heavy load.
Sacral (S1–S5)	Five vertebrae fused to form the sacrum; triangular shape; sacral canal; anterior and posterior sacral foramina; smooth anterior surface and rough posterior surface.	Transfers body weight from the spine to the pelvis and provides stability to the pelvic girdle. Limited movement for structural support.
Coccygeal	Usually four fused rudimentary vertebrae; lack pedicles, laminae and vertebral foramina; small and tapered structure.	Provides attachment for pelvic floor muscles and ligaments; minor role in weight support when sitting.



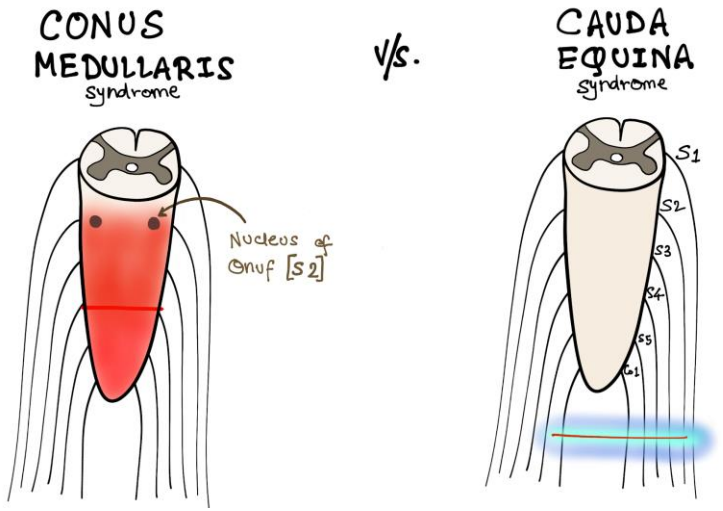
Relation of Nerve Roots to Vertebrae

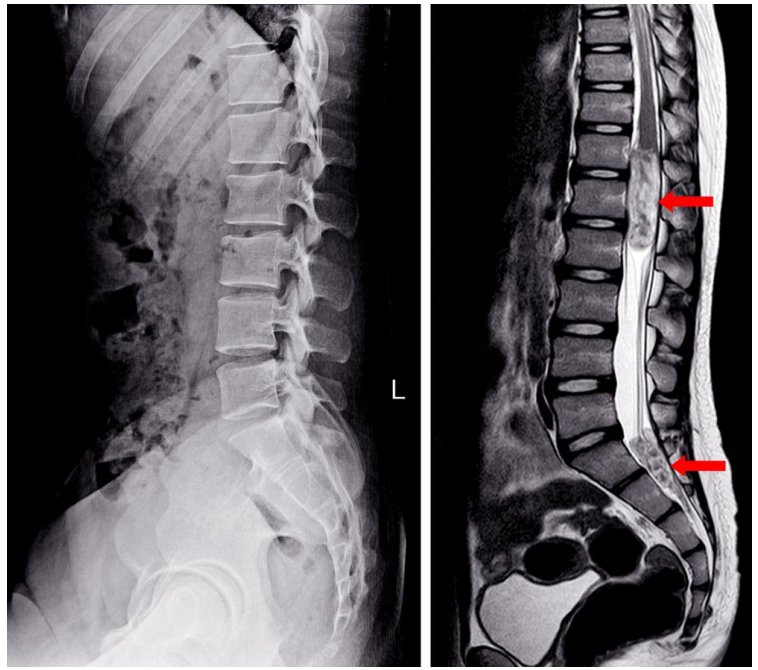
Region	Relationship between Nerve Roots and Vertebrae
Cervical (C1–C7)	Nerve roots exit above their corresponding vertebrae
C8 Nerve Root	Exits below C7 vertebra and above T1
Thoracic (T1–T12)	Nerve roots exit below their corresponding vertebrae
Lumbar (L1–L5)	Nerve roots descend and exit well below their vertebral level
Sacral (S1–S5)	Nerve roots descend within the sacral canal before exiting via sacral foramina



Conus Medullaris Syndrome

- The conus medullaris is the terminal end of the spinal cord, which typically occurs at the L1 vertebral level in the average adult.
- Conus medullaris syndrome (CMS) results when there is compressive damage to the spinal cord from T12-L2.
- The cauda equina is a group of nerves and nerve roots stemming from the distal end of the spinal cord, typically levels L1-L5 and contains axons of nerves that give both motor and sensory innervation to the legs, bladder, anus, and perineum.
- Differentiating conus medullaris syndrome from cauda equina syndrome is clinically crucial.
- Conus medullaris syndrome presents with more symmetrical deficits, early sphincter dysfunction, and less severe radicular pain. Damage here can result in both UMN (upper motor neuron) and LMN (lower motor neuron) symptoms, including weakness, loss of sensation, and bladder or bowel dysfunction.
- Cauda Equina Injury: Below the conus medullaris, the nerve roots form the cauda equina. A herniated disc at the same level can compress these nerve roots, leading to LMN signs such as muscle weakness, sensory deficits, and, often, severe pain or loss of function in the legs and pelvic organs.

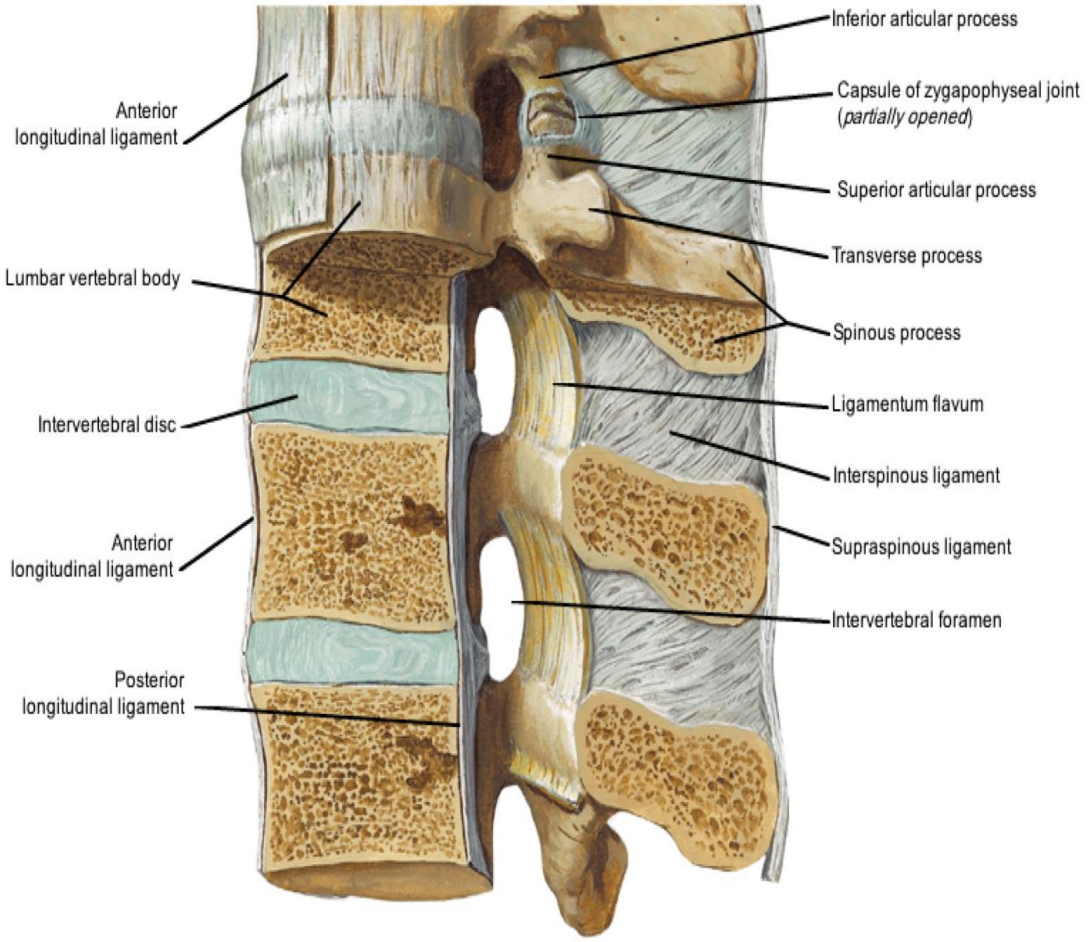


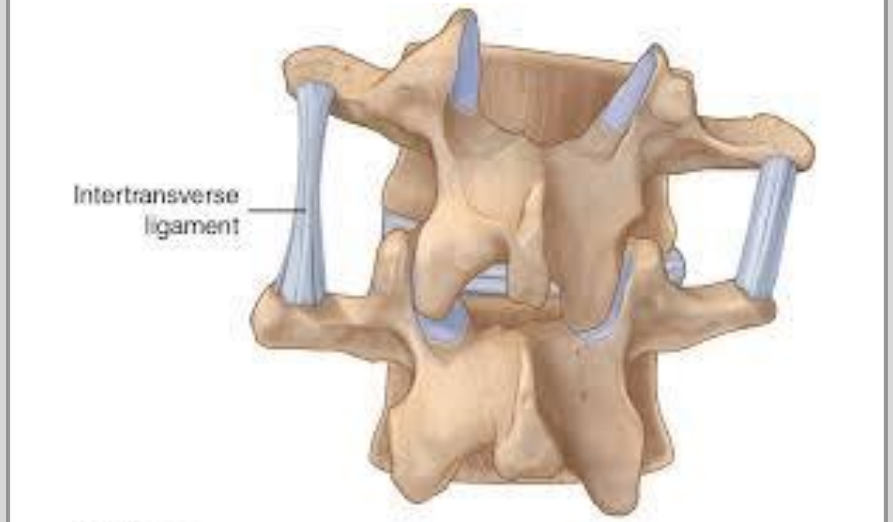


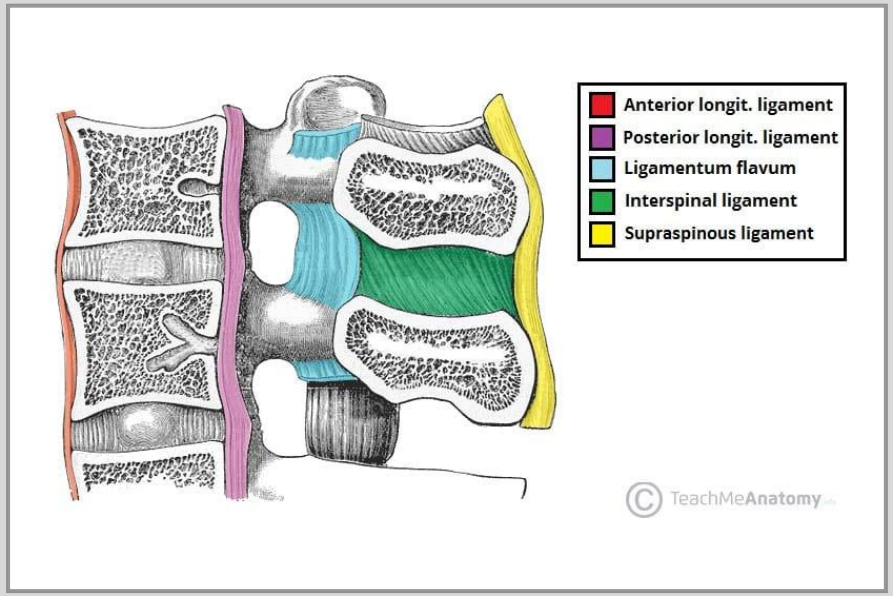
Conus Medullaris and Cauda Equina injury

A single disc herniation at the L1-L2 or adjacent levels can cause both conus medullaris and cauda equina syndrome because it can compress both the conus medullaris itself and the cauda equina nerve roots simultaneously.

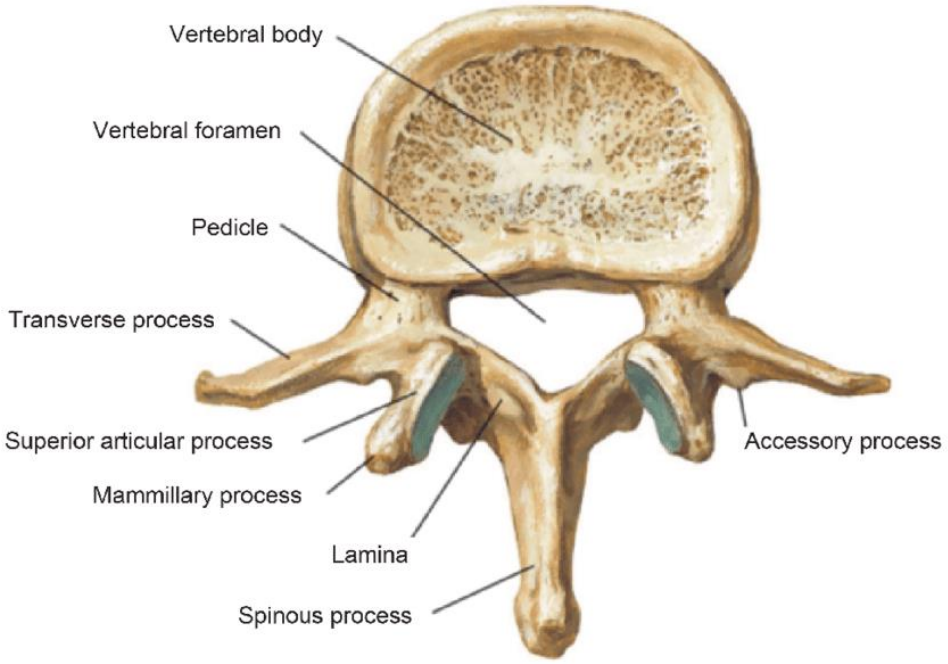
This results in a combination of upper motor neuron and lower motor neuron signs, along with severe sensory and motor deficits, as well as bladder and bowel issues.


Inferior articular process
Capsule of zygapophyseal joint (*partially opened*)
Superior articular process
Transverse process
Spinous process
Ligamentum flavum
Interspinous ligament
Supraspinous ligament
Intervertebral foramen
Anterior longitudinal ligament
Lumbar vertebral body
Intervertebral disc
Anterior longitudinal ligament
Posterior longitudinal ligament

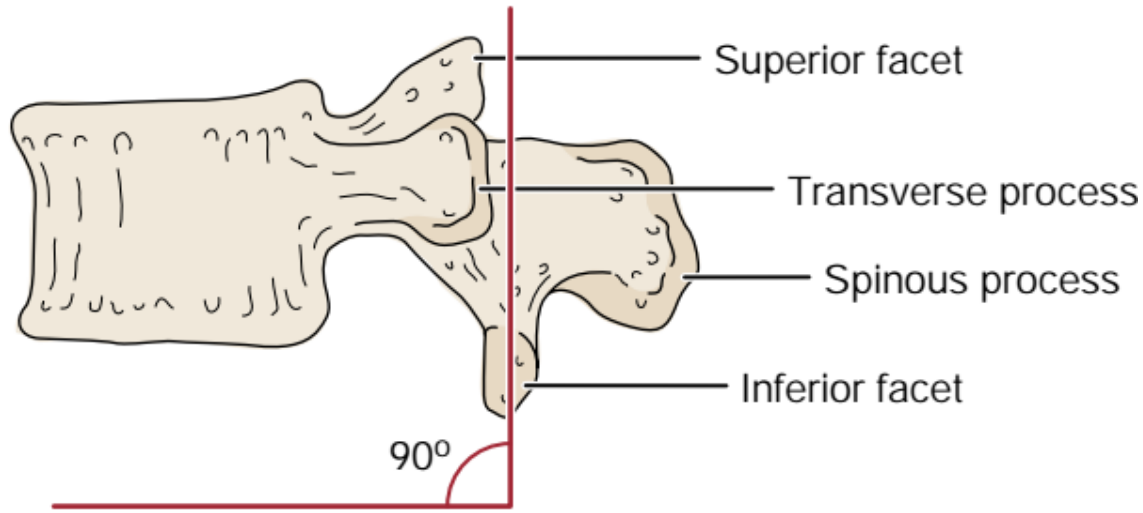

Intertransverse ligament


Anterior longit. ligament
Posterior longit. ligament
Ligamentum flavum
Interspinal ligament
Supraspinous ligament
© TeachMeAnatomy

Anatomy of the lumbar spine



L2 vertebra:
superior view



Assessment of Lumbosacral Region

1- Subjective Assessment: History taking, clearing red flags, planning objective Assessment

2- Objective Assessment: Observation, palpation, ROM, MMT, neurological Ax, Gait Ax, Functional Ax, Special testing

Clearing Red flags

Multiple Myeloma or Metastatic Spinal Cord Compression (MSCC)

Spinal Infection

Osteoporotic Vertebral Compression Fracture

Cauda Equina Syndrome (CES)

Criteria for recognising red flags

Myelopathy

- Insidious progression
- Gait disturbance, ataxia
- Clumsy or weak hands
- Combination of neuro changes in upper and lower limbs
- +ve Hyperreflexia
- +ve Clonus
- +ve Babinski sign
- +ve L'hermitte's sign
- +ve Hoffman's sign

compression on spinal cord

- Altered sensation below a certain level or hemisensory loss;
- Motor symptoms of hemiplegia/ hemiparesis, paraplegia/paraparesis or tetraplegia/tetraparesis; and/or autonomic symptoms including constipation and urinary retention.
- Altered reflexes. Hyper-reflexia and loss of pinprick sensation, temperature, position, and vibratory sensation may occur early, especially when associated with malignancy.

Cancer/Tumour

- Atypical mass / swelling with or without the presence of pain
- Pain with no history of injury
- Unremitting or increasing pain
- Weight loss
- Past medical history of Cancer
- Systemically unwell
- Night pain and disturbed sleep

Infection/inflammation

- Red, hot, swollen, painful
- Wound leakage
- Systematically unwell, malaise
- Fever
- Gross restriction of movement
- Sever pain with no history of trauma
- Night pain
- Discitis, sudden onset of acute spinal pain or change in pattern with combination of above.

Criteria for recognising red flags

Inflammatory Arthropathy

- Morning stiffness & backache, or multiple joint problems (pain / stiffness / swelling). No improvement with rest.
- Generally unwell
- Associated skin rash, inflammatory bowel disease, eye problems, (uveitis / conjunctivitis), urethritis
- Sacroiliac pain / tenderness
- Altered blood results. Positive C-Reactive protein (CRP), ESR, Plasma viscosity (PV), +ve HLA B27 test

Aortic Aneurysm

•Over 60
•Acute, sudden onset of back pain
•Low back pain that is severe - doubled over in pain
•Severe abdominal pain
•Continuous pain, no better with rest
•Pain may radiate into the groin or leg
•Pain may be accompanied by symptoms of internal bleeding, such as nausea, vomiting, rapid heart rate, cool or clammy skin, sweating, and / or shortness of breath

DVT

•Oedema at calf
•Red, hot limb
•Painful -often behind the knee but not in all cases
•Feet cyanosis deep red/purple
•Well's Criteria

Wells' Criteria for DVT (free access score system)

Active cancer Treatment or palliation within 6 months	**No 0**	Yes +1
Bedridden recently >3 days or major surgery within 12 weeks	**No 0**	Yes +1
Calf swelling >3 cm compared to the other leg Measured 10 cm below tibial tuberosity	**No 0**	Yes +1
Collateral (nonvaricose) superficial veins present	**No 0**	Yes +1
Entire leg swollen	**No 0**	Yes +1
Localized tenderness along the deep venous system	**No 0**	Yes +1
Pitting edema, confined to symptomatic leg	**No 0**	Yes +1
Paralysis, paresis, or recent plaster immobilization of the lower extremity	**No 0**	Yes +1
Previously documented DVT	**No 0**	Yes +1
Alternative diagnosis to DVT as likely or more likely	**No 0**	Yes -2

History taking and patient interview

Patient’s age, occupation, mechanism of injury?

How long has the problem bothered the patient?

Is the pain deep? Superficial? Shooting? Burning? Aching?

Is the pain improving? Worsening? Staying the same?

Is there any increase in pain with coughing? Sneezing? Deep breathing? Straining? Laughing?

Are there any postures or actions that specifically increase or decrease the pain or that cause difficulty?

Is the pain worse in the morning or evening? Does the pain get better or worse as the day progresses? Does the pain wake you up at night?

Which movements hurt? Which movements are stiff?

Has the patient noticed any weakness or decrease in strength? Has the patient noticed that his or her legs have become weak while walking or climbing stairs?

What is the patient’s usual activity or pastime? Before the injury, did the patient modify or perform any unusual repetitive or high-stress activity?

What is the patient’s sleeping position? Does the patient have any problems sleeping? What type of mattress does the patient use (hard, soft)?

Does the patient have any difficulty with micturition?

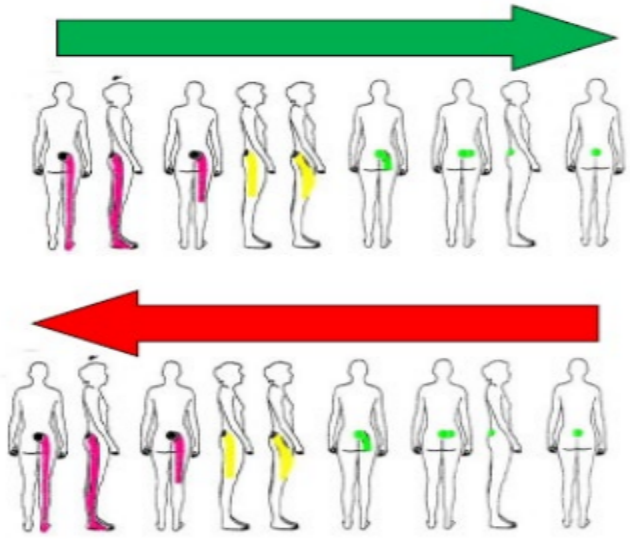
Specific Questions

- Dizziness/Dysarthria/Dysphagia/Diplopia/Drop attacks (5D's)
- Nausea/Nystagmus/Numbness face/tongue (3N's)
- Clumsy hands
- Gait disturbance
- Lhermitte's sign
- Bilateral P&N
- Hormone replacement therapy (HRT)
- Long term cortisone use
- Contraceptive pill
- Recent dental surgery
- TB/Cancer (breast/lung/prostate/thyroid/kidney)
- Osteoporosis familial /DEXA Scan
- CES symptoms questioning
- Cardiovascular history
- Inflammatory history
- Recent infection and new onset LBP
- Cough / sneeze / strain

History taking and patient interview

Is there any radiation of pain? Is the pain centralizing or peripheralizing ?

Pain originating from the back can sometimes travel up and down the leg. Pain that spreads further down the legs is described as peripheralization. Pain that recedes towards the back to a smaller, more local, area, is described as centralization. This is caused by a change in the stress and strain on the joints, soft tissues, and nerves as they exit the spine.



Centralization

Centralization of pain that occurs with specific positions or movements **is desirable.** This indicates that there is reduced stress and strain on the joints, soft tissues, and nerves of the back. Use these positions or movements that centralize pain to help get better.

Peripheralization

Peripheralization of pain that occurs during certain positions or movements is **NOT desirable**. This indicates that there is increased stress and strain on the joints, soft tissues, and nerves of the back. Avoid activities that cause peripheralization of symptoms.

In the beginning, some back pain is normal with exercise as long as pain is receding out of the leg or the area of back pain is getting smaller, this is a good prognostic sign (centralization). Stop any exercise that makes the pair travel further down the leg.

The Keele STaRT Screening Tool

The © Keele STaRT Back Screening Tool

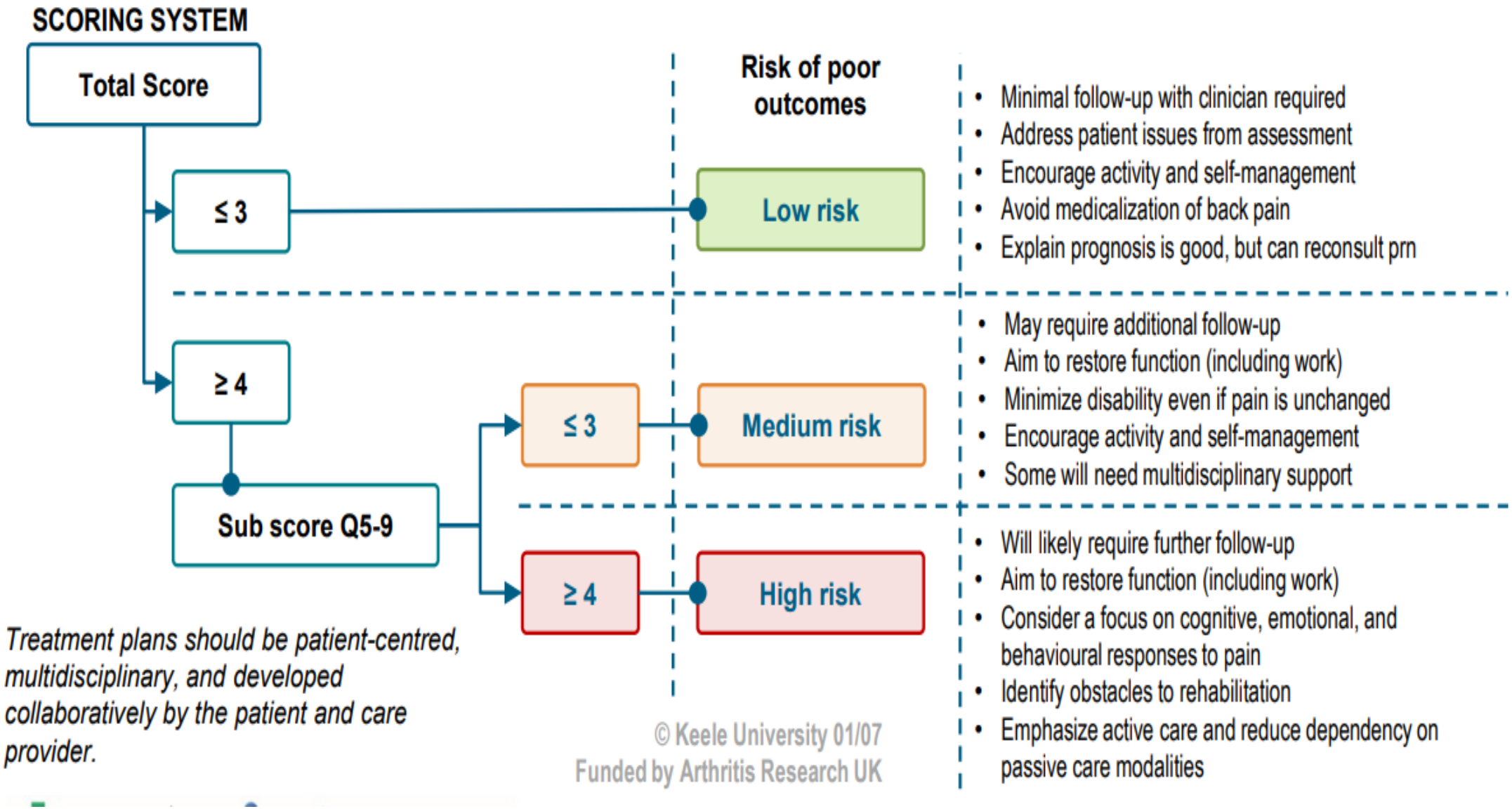
This tool characterizes patients by risk of persistent symptoms (low, medium, or high) which assesses likely recovery outcomes and allows the clinician to tailor interventions appropriately

Thinking about the last 2 weeks, "X" your response to the following questions:	Disagree (0)	Agree (1)
1. My back pain has **spread down my leg(s)** at some time in the last 2 weeks		
2. I have had pain in the **shoulder** or **neck** at some point in the last 2 weeks		
3. I have only **walked short distances** because of my back pain		
4. In the last 2 weeks, I have **dressed more slowly** than usual because of back pain		
5. It's not really safe for a person with a condition like mine to be physically active		
6. Worrying thoughts have been going through my mind a lot of the time		
7. I feel that **my back pain is terrible** and **it's never going to get any better**		
8. In general, I have **not enjoyed** all the things I used to enjoy		

9. Overall, how bothersome has your back pain been in **the last 2 weeks**?

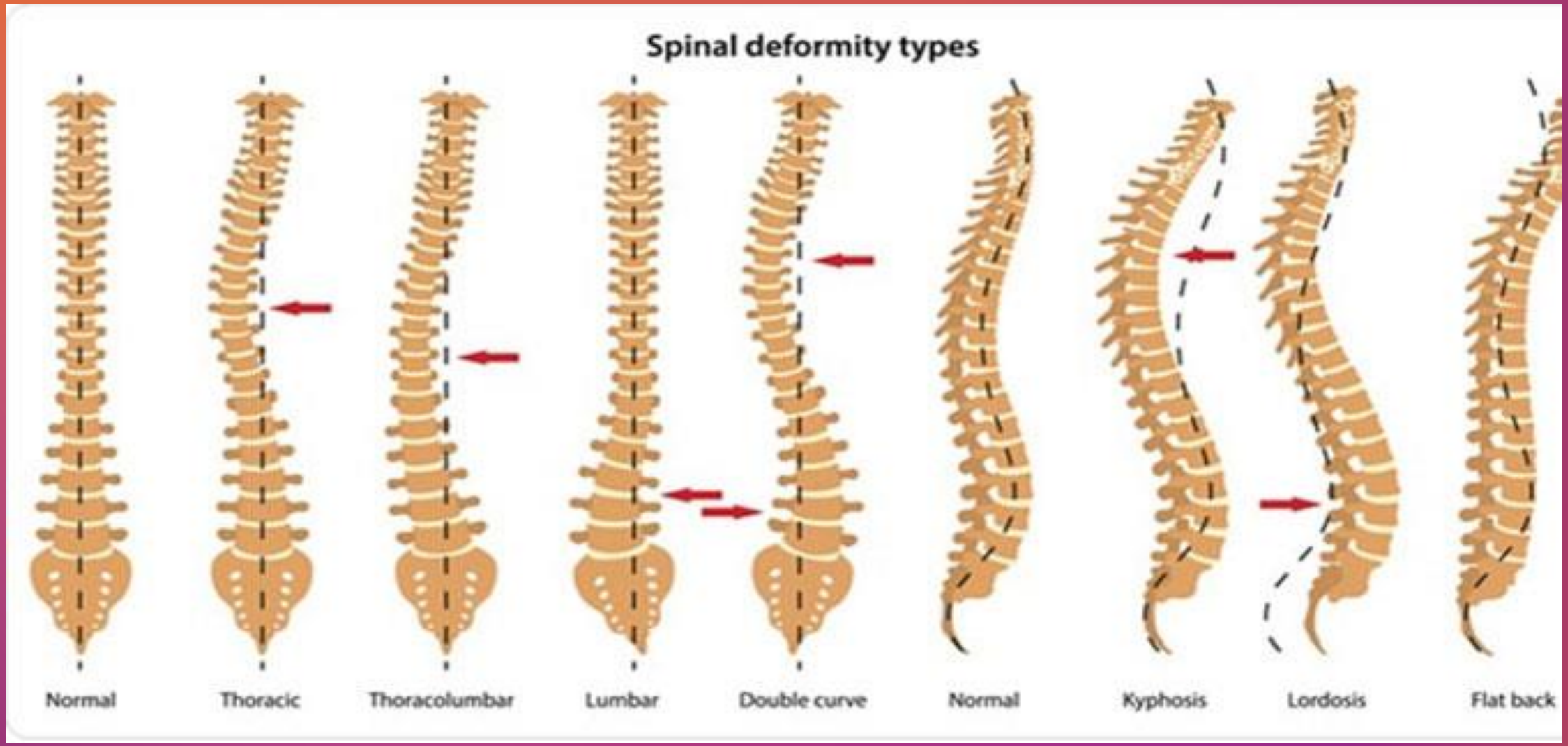
Not at all	Slightly	Moderately	Very much	Extremely
☐	☐	☐	☐	☐
0	**0**	**0**	**1**	**1**

TOTAL score (sum all 9): _____ SUB score (sum of Q5-9): _____


SCORING SYSTEM
Total Score
≤ 3
≥ 4
Sub score Q5-9
≤ 3
≥ 4
Risk of poor outcomes
Low risk
Medium risk
High risk
• Minimal follow-up with clinician required
• Address patient issues from assessment
• Encourage activity and self-management
• Avoid medicalization of back pain
• Explain prognosis is good, but can reconsult prn
• May require additional follow-up
• Aim to restore function (including work)
• Minimize disability even if pain is unchanged
• Encourage activity and self-management
• Some will need multidisciplinary support
• Will likely require further follow-up
• Aim to restore function (including work)
• Consider a focus on cognitive, emotional, and behavioural responses to pain
• Identify obstacles to rehabilitation
• Emphasize active care and reduce dependency on passive care modalities
Treatment plans should be patient-centred, multidisciplinary, and developed collaboratively by the patient and care provider.
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Funded by Arthritis Research UK

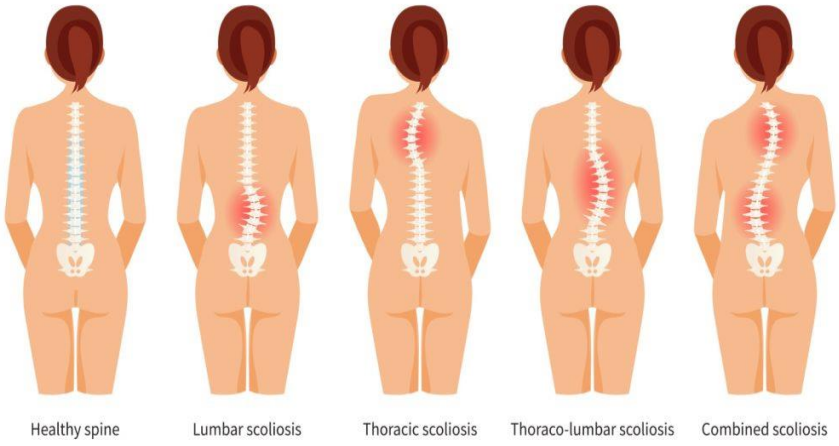
Lumbar spine Objective assessment

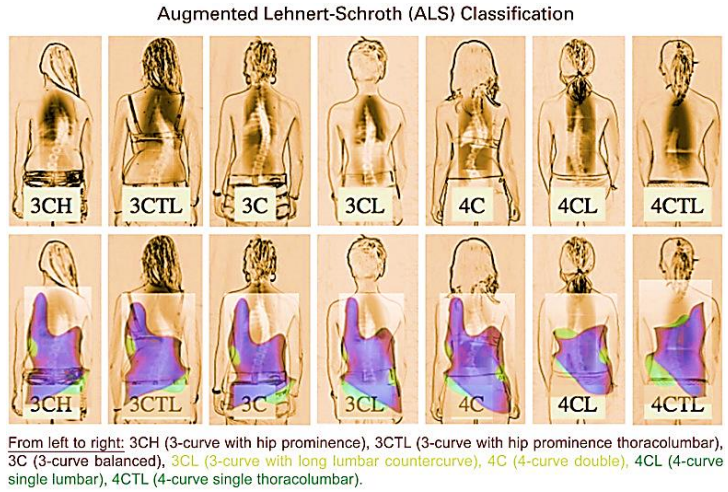
Observation

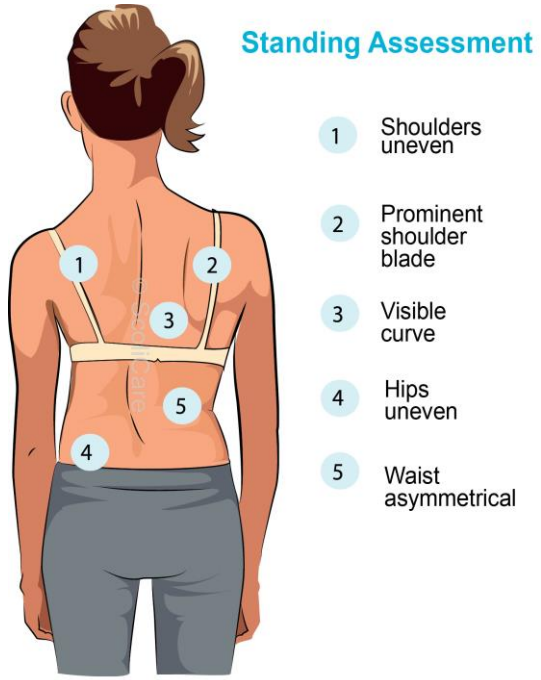


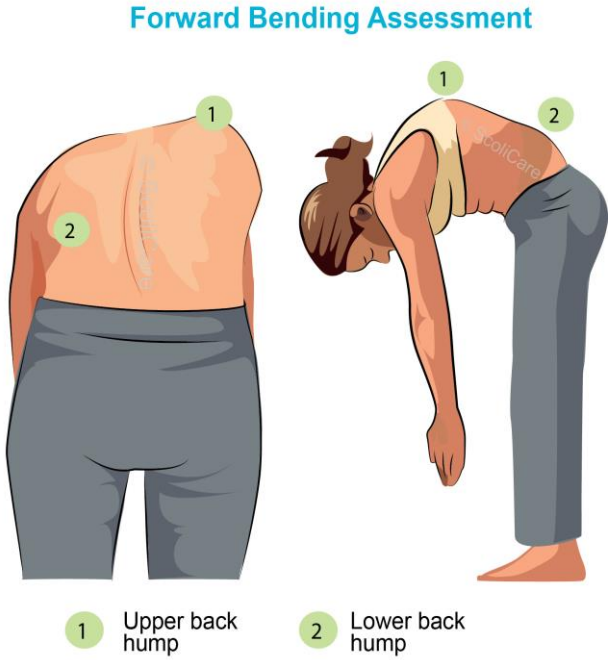
Observation

Scoliosis is a multifactorial three-dimensional (3D) spinal deformity that always involves elemental deformities in the three main planes: lateral curvature in frontal, anteroposterior (mainly lordotic) deviation in sagittal, and (very characteristically) vertebral axial rotation in the horizontal plane.



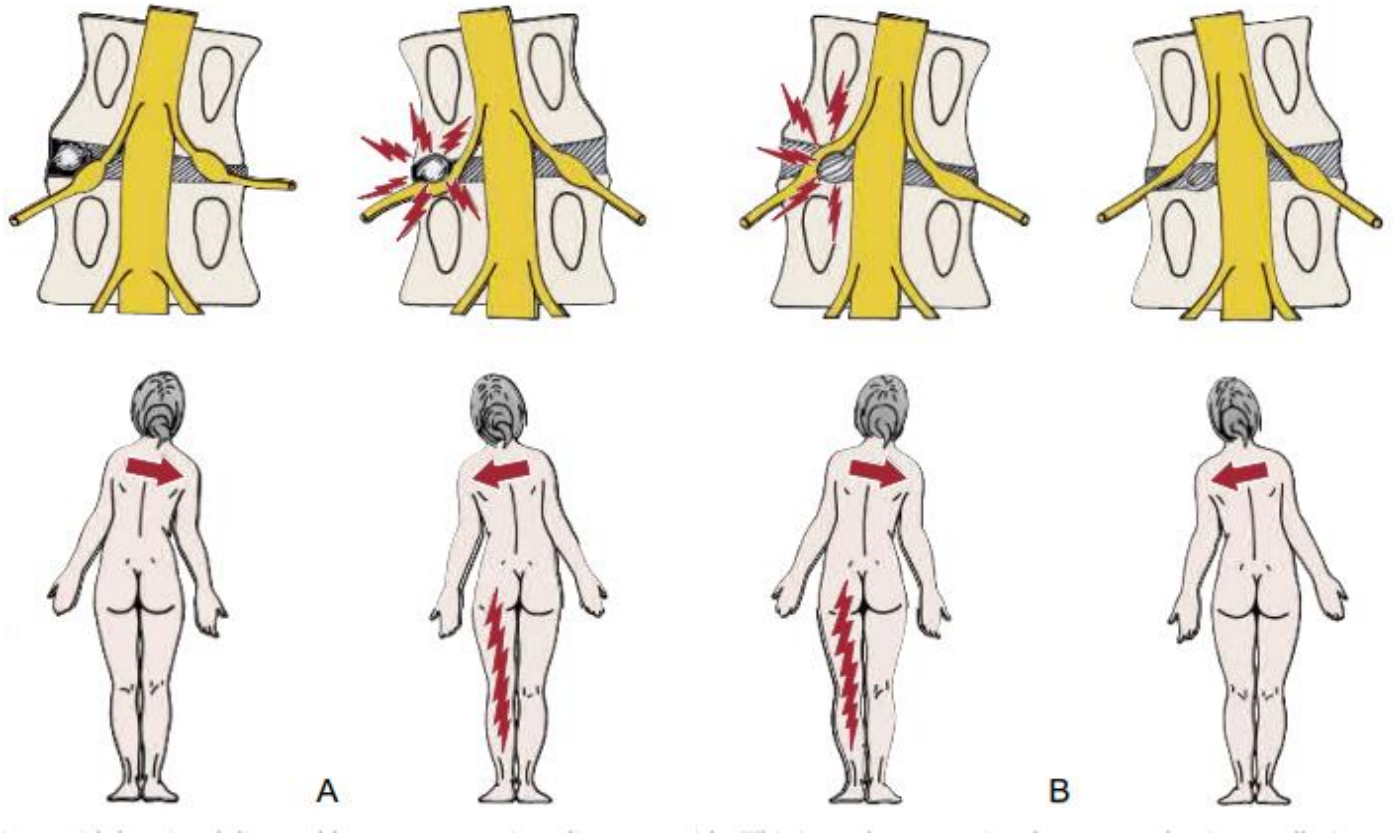






Lumbar list, Lateral Shift or antalgic postural shift

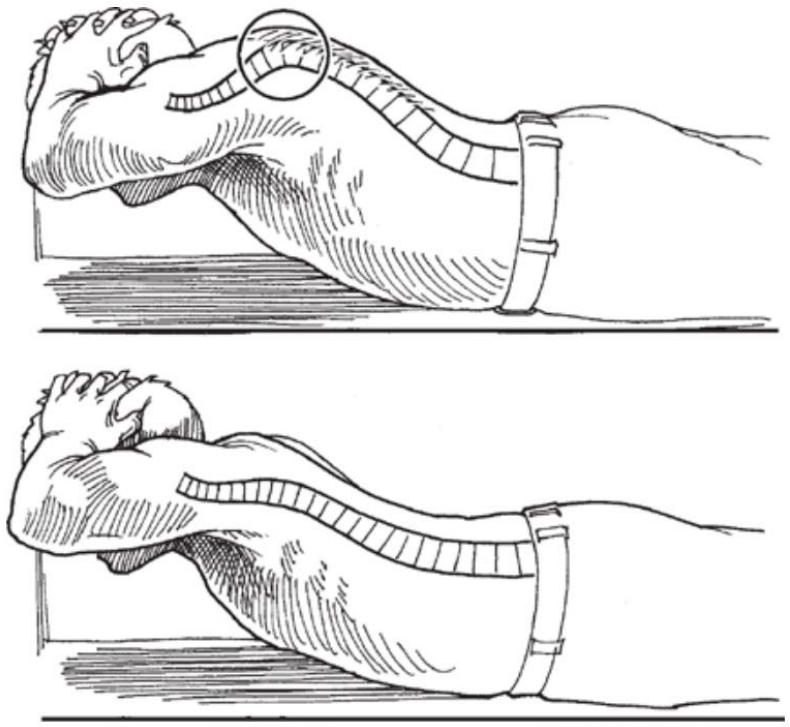


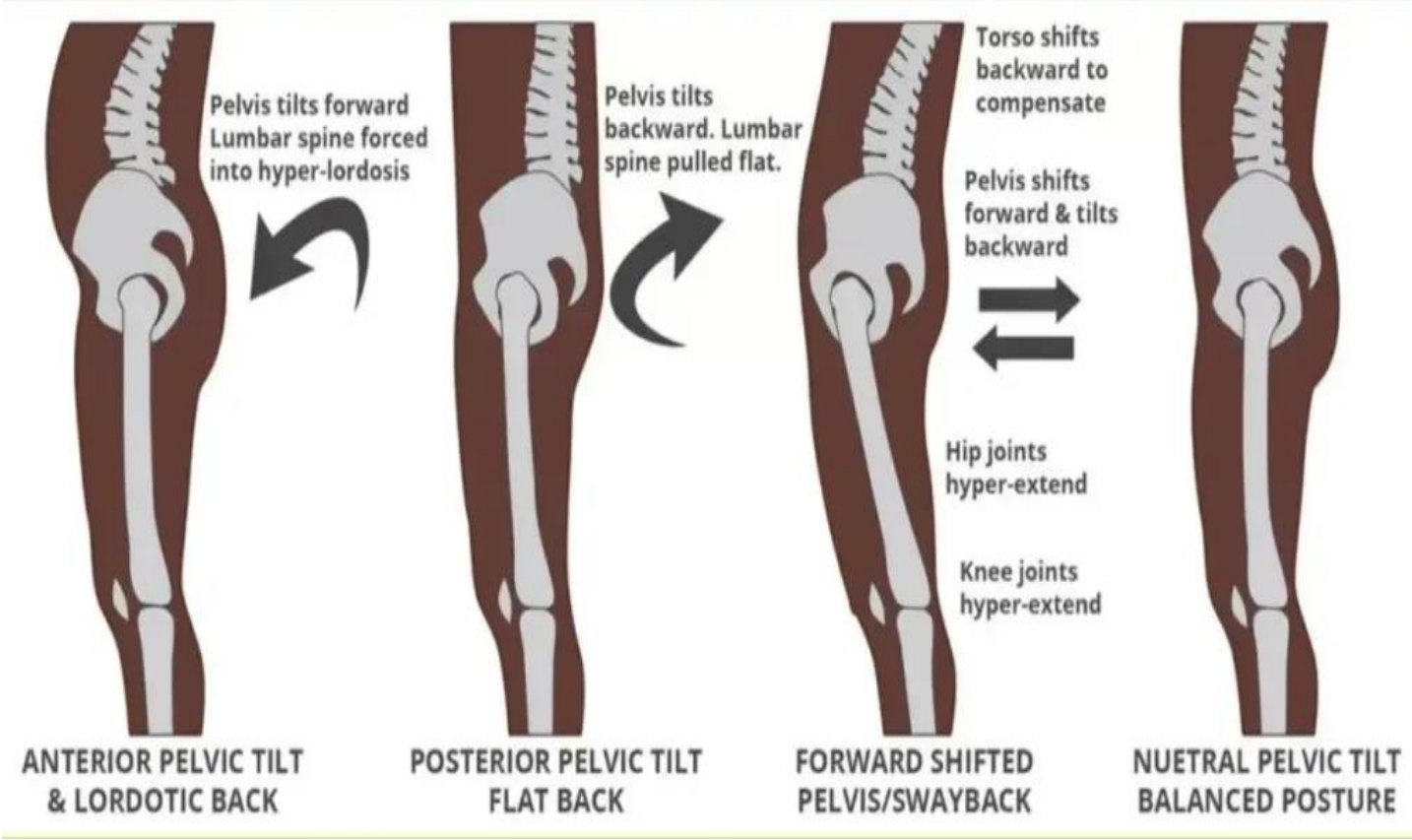


- Patients with herniated disc problems may sometimes list to one side. This is a voluntary or involuntary mechanism to alleviate nerve root irritation.
- The list in some patients is toward the side of the sciatica; in others, it is toward the opposite side. A reasonable hypothesis suggests that when the herniation is lateral to the nerve root (A), the list is to the side opposite the sciatica because a list to the same side would elicit pain.
- Conversely, when the herniation is medial to the nerve root (B), the list is toward the side of the sciatica because tilting away would irritate the root and cause pain.

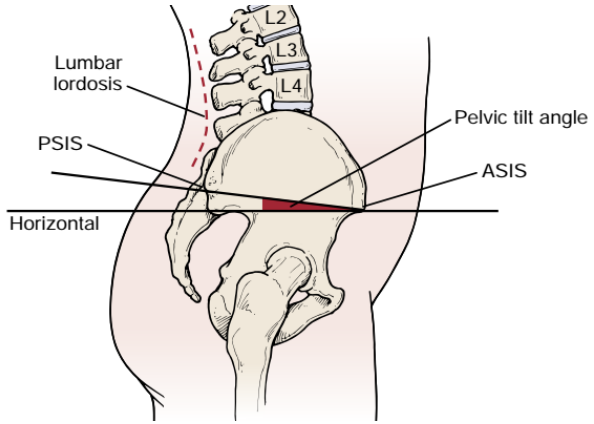
(Redrawn from White AA, Panjabi MM: Clinical biomechanics of the spine, ed 2, Philadelphia, 1990, JB Lippincott, p 415.) (© Augustus A. White III and MM Panjabi.)

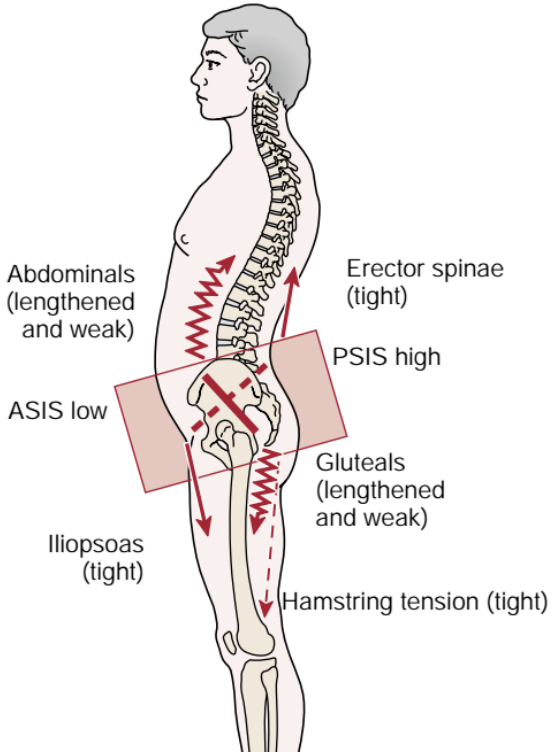
Structural and mobile kyphosis – Lumbar Postural anomalies

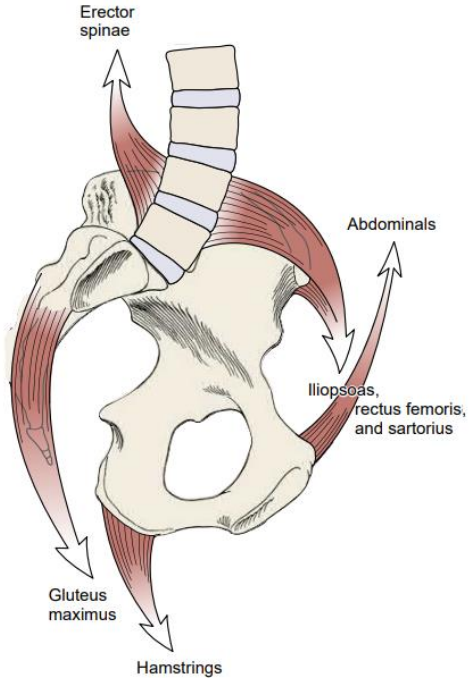


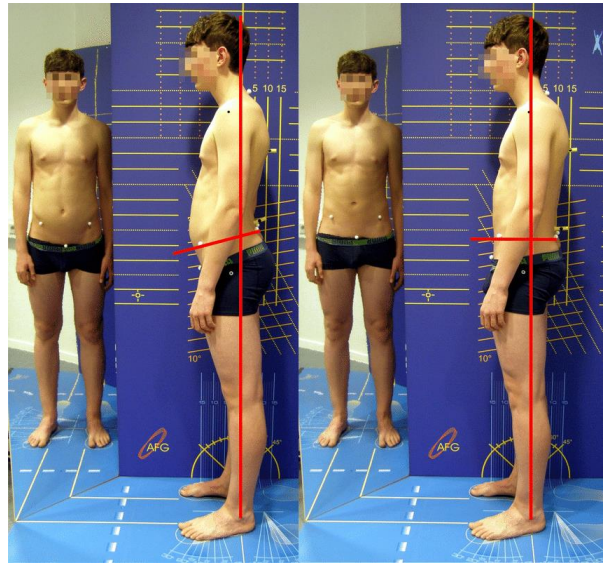


Pelvic Tilt angle (7-15 degrees)









ROM Ax

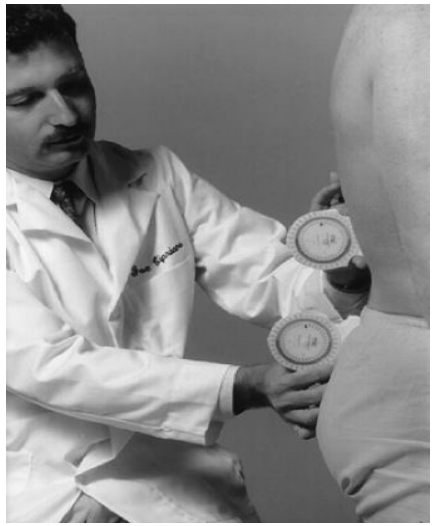
Forward flexion (40°–60°)
Extension (20°–35°)
Side (lateral) flexion, left and right (15°–20°)
Rotation, left and right (3°–18°)

Modified Schober's test :

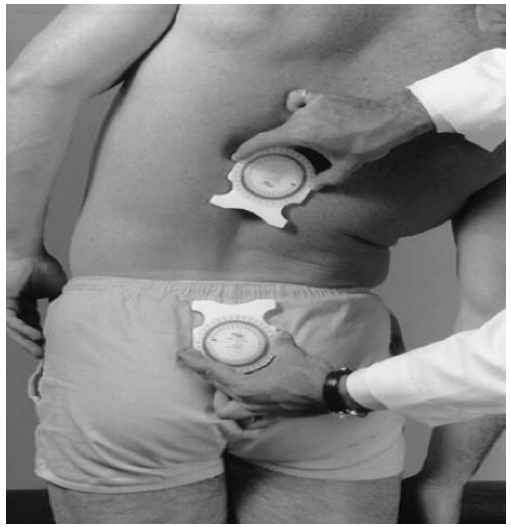
- Mark the midpoint between 2 posterior superior iliac spines (L5-S2).
- Mark another point 10 cm above and a third point 5 cm below.
- In neutral standing, measure the distance between the upper and lower marks.
- The patient is then asked to bend forward as far as possible for lumbar flexion.
- normal increase of distance typically being around 5–7 cm.
- For extension, the reduction in distance is recorded, which normally decreases by about 1–2 cm.

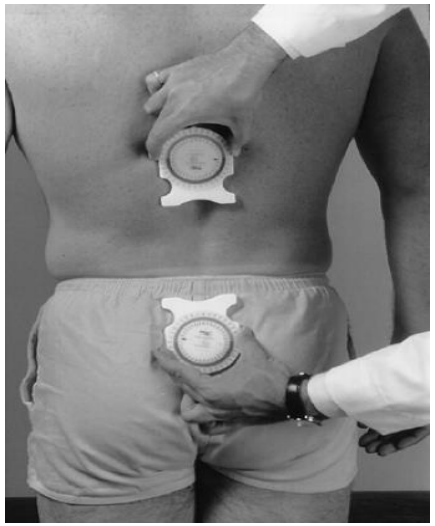








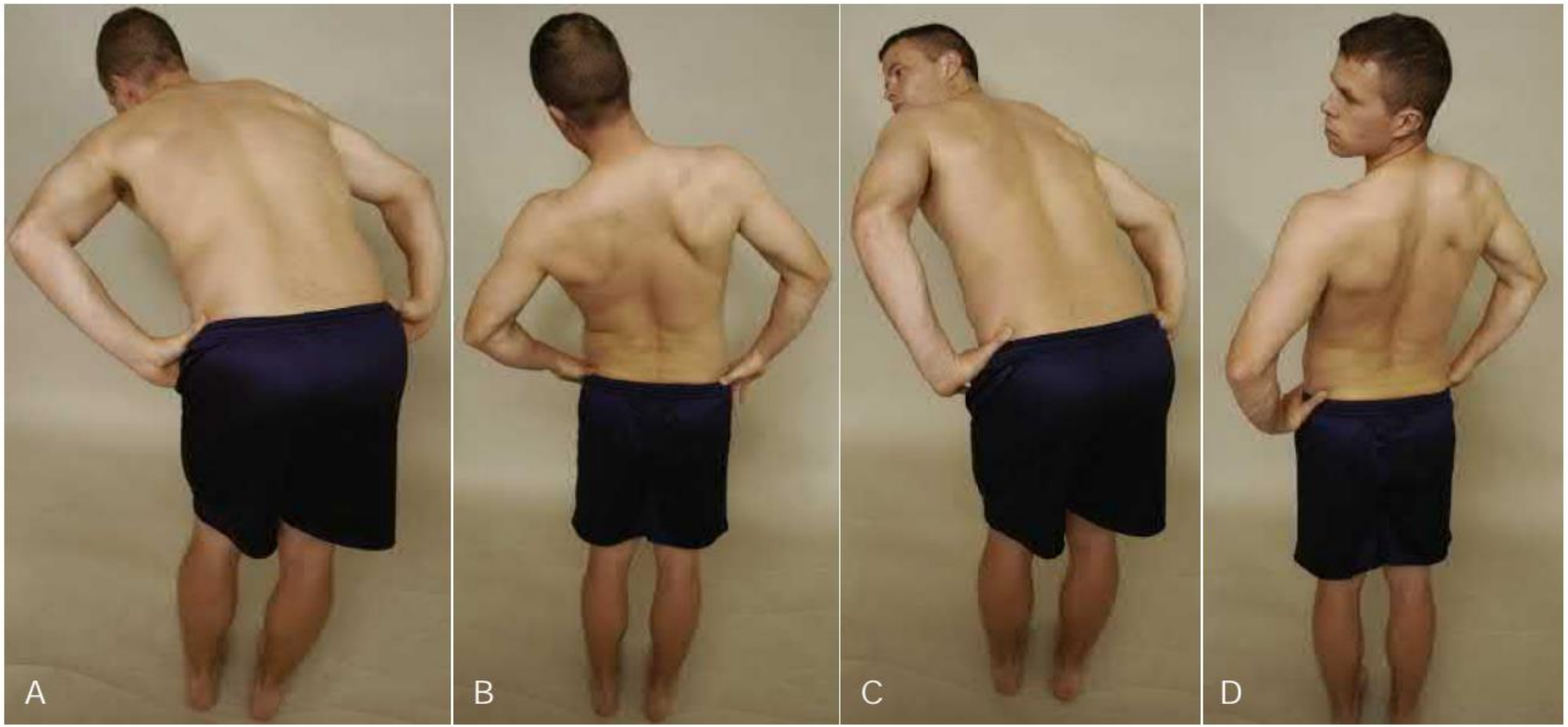




- With the patient standing in neutral, place one inclinometer flat at the T12 spinous process in the coronal plane and the second inclinometer at S2, also in the coronal plane.
- Zero out both inclinometers. Instruct the patient to flex the trunk to one side.
- Record the inclination of both inclinometers.
- Subtract the sacral inclination from the T12 inclination to obtain the lumbar lateral flexion angle.
- Perform measurements for both right and left lateral flexion.



Combined movement



Accessory movement for the lumbar spine

Joint play movements of the lumbar spine:

- (A) Flexion.
- (B) Extension.
- (C) Side flexion.
- (D) Posteroanterior central vertebral pressure.
- (E) Posteroanterior unilateral vertebral pressure.
- (F) Transverse vertebral pressure.

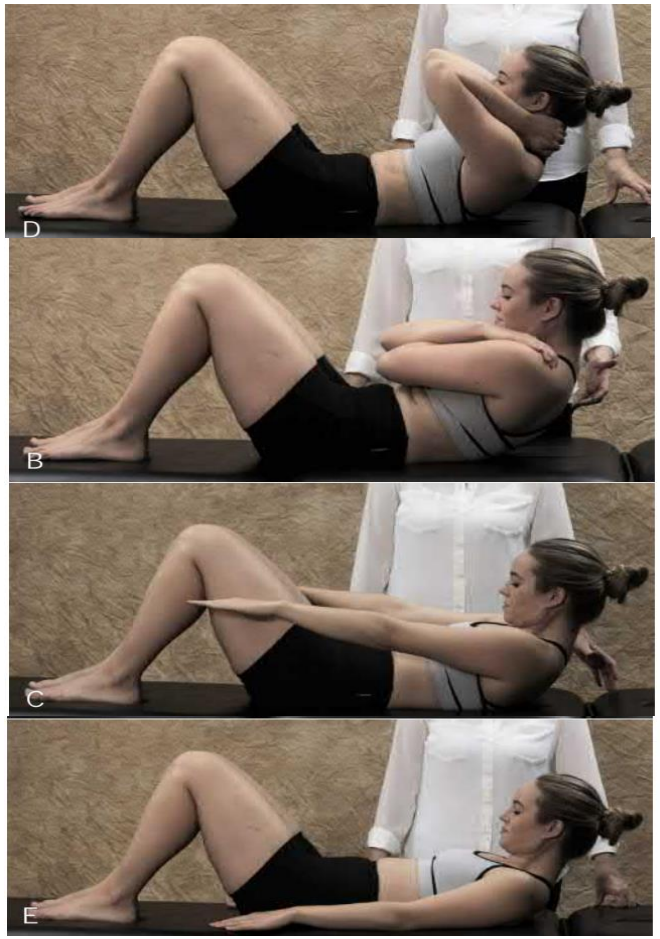
Manual Muscle testing MMT

Muscles of the Lumbar Spine: Their Actions and Nerve Root Derivations

Action	Muscles Acting	Nerve Root Derivation
Forward flexion	1. Psoas major	L1–L3
	2. Rectus abdominis	T6–T12
	3. External abdominal oblique	T7–T12
	4. Internal abdominal oblique	T7–T12, L1
	5. Transversus abdominis	T7–T12, L1
	6. Intertransversarii	L1–L5
Extension	1. Latissimus dorsi	Thoracodorsal (C6–C8)
	2. Erector spinae	L1–L3
	Iliocostalis lumborum	L1–L3
	Longissimus thoracis	L1–L5
	3. Transversospinalis	L1–L5
	4. Interspinales	L1–L5
	5. Quadratus lumborum	T12, L1–L4
	6. Multifidus	L1–L5
	7. Rotatores	L1–L5
	8. Gluteus maximus	L1–L5
Side flexion	1. Latissimus dorsi	Thoracodorsal (C6–C8)
	2. Erector spinae	L1–L3
	Iliocostalis lumborum	L1–L3
	Longissimus thoracis	L1–L5
	3. Transversalis	L1–L5
	4. Intertransversarii	L1–L5
	5. Quadratus lumborum	T12, L1–L4
	6. Psoas major	L1–L3
	7. External abdominal oblique	T7–T12
Rotation[a]	1. Transversalis	L1–L5
	2. Rotatores	L1–L5
	3. Multifidus	L1–L5

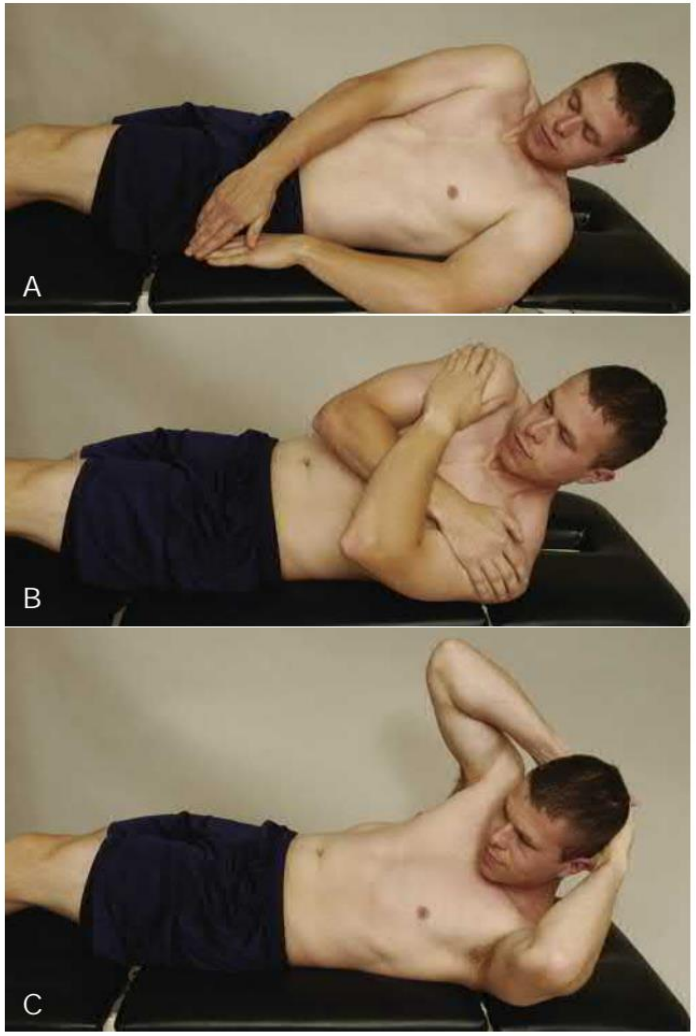
Resisted isometric movements of the lumbar spine. (A) Flexion, extension, and side flexion. (B) Rotation to right.





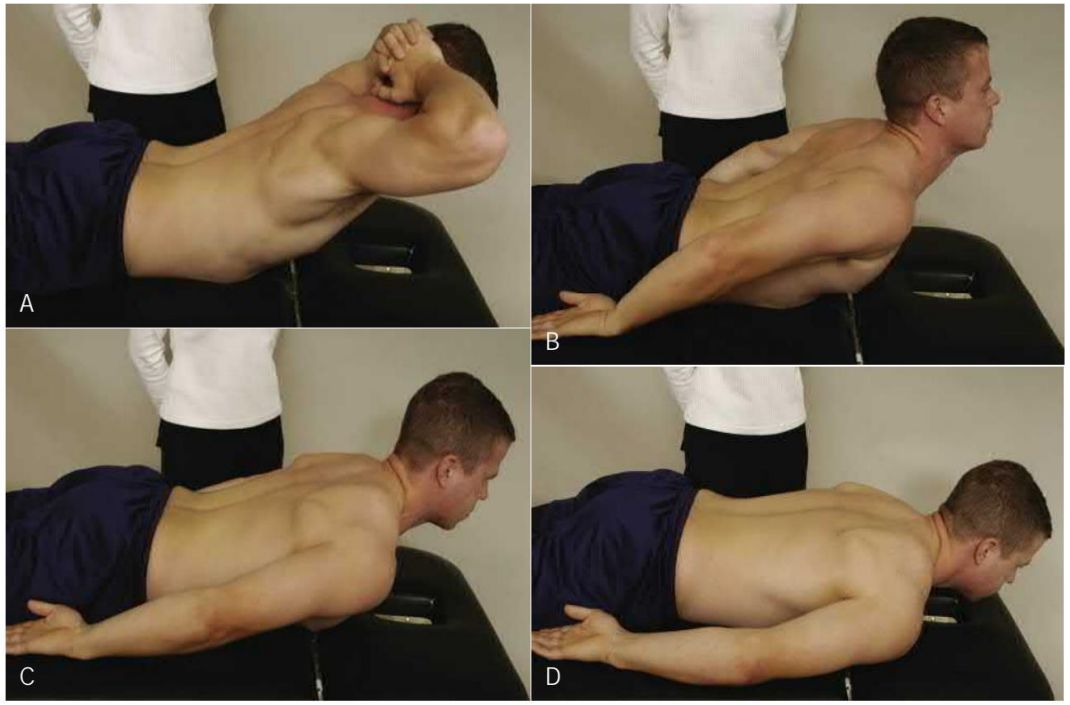
MMT of the lumbar spine (Isometric abdominal test)

- Normal (5) = Hands behind neck until scapulae clear table (20- to 30-second hold)
- Good (4) = Arms crossed over chest until scapulae clear table (15- to 20-second hold)
- Fair (3) = Arms straight until scapulae clear table (10-to 15-second hold)
- Poor (2) = Arms extended toward knees until tops of scapulae lift from table (1- to 10-second hold)
- Trace (1) = Unable to raise more than head off table



Internal/external abdominal oblique Isometric test

- Normal (5) = Flexes and rotates the lumbar spine fully with hands behind head (20- to 30-second hold)
- Good (4) = Flexes and rotates the lumbar spine fully with hands across chest (15- to 20-second hold)
- Fair (3) = Flexes and rotates the lumbar spine fully with arms reaching forward (10- to 15-second hold)
- Poor (2) = Unable to flex and rotate fully (01- to 10-second hold)
- Trace (1) = Only slight contraction of the muscle with no movement
- (0) = No contraction of the muscle.



Isometric Extensors Test

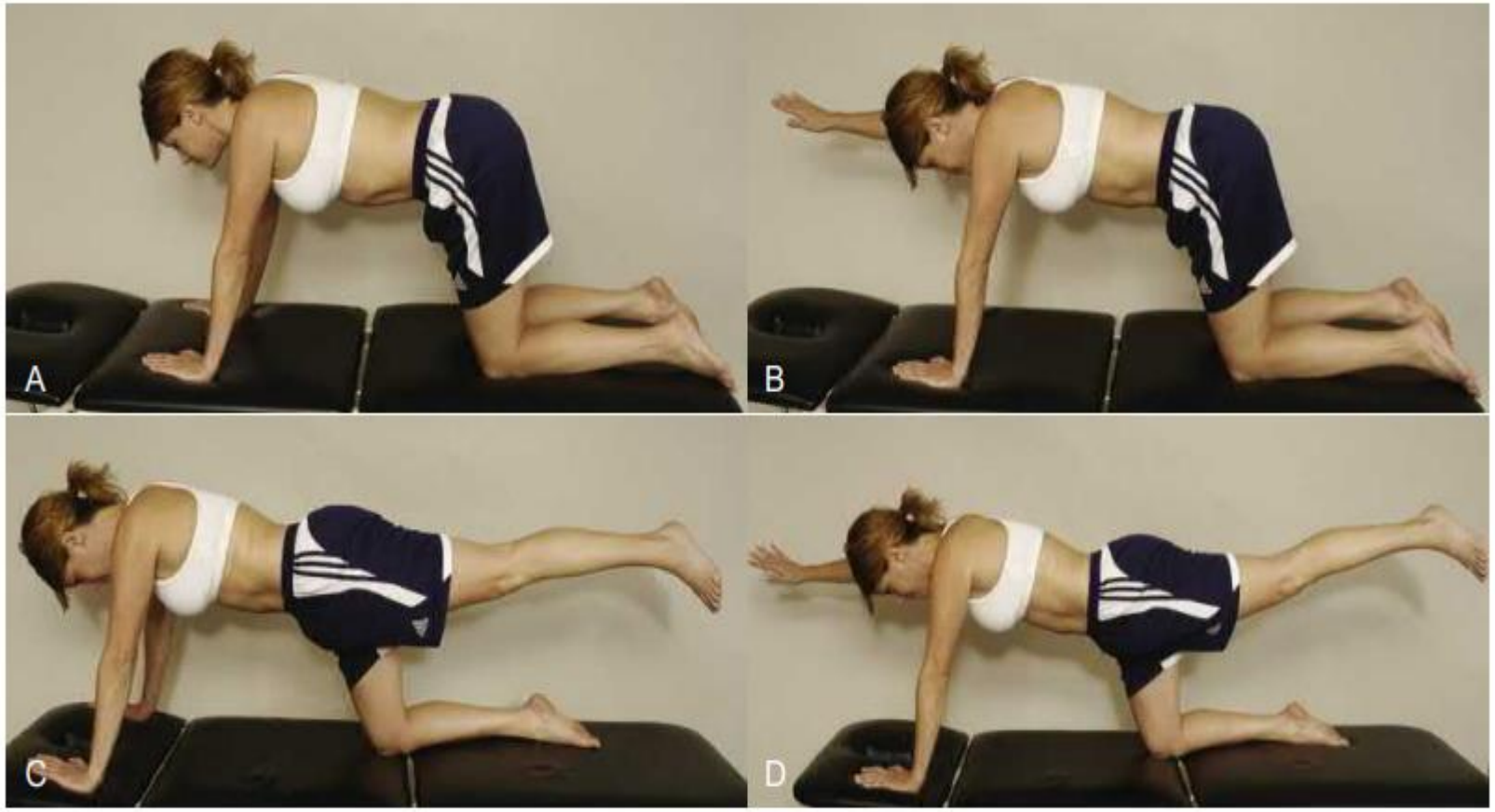
- Normal (5) = With hands clasped behind the head, extends the lumbar spine, lifting the head, chest, and ribs from the floor (20- to 30-second hold)
- Good (4) = With hands at the side, extends the lumbar spine, lifting the head, chest, and ribs from the floor (15- to 20-second hold)
- Fair (3) = With hands at the side, extends the lumbar spine, lifting the sternum off the floor (10- to 15-second hold)
- Poor (2) = With hands at the side, extends the lumbar spine, lifting the head off the floor (1- to 10-second hold)
- Trace (1) = Only slight contraction of the muscle with no movement
- (0) = No contraction of the muscle.



Dynamic Horizontal Side Support (Side Bridge) Test (Quadratus Lumborum)

- Normal (5) = Able to lift pelvis off examining table and hold spine straight (10- to 20-second hold) .
- Good (4) = Able to lift pelvis off examining table but has difficulty holding spine straight (5- to 10-second hold)
- Fair (3) = Able to lift pelvis off examining table and cannot hold spine straight
- Poor (2) = Unable to lift pelvis off examining table
- Trace (1) = Only slight contraction of the muscle with no movement
- (0) = No contraction of the muscle.

Back rotators test (Multifidus)



1. Single straight arm lift and hold
2. Single straight leg lift and hold
3. Contralateral straight arm and straight leg lift and hold The scoring for the test would be as follows:

• Normal (5) = Able to do contralateral arm and leg, both sides, while maintaining neutral pelvis (20- to 30-second hold)

• Good (4) = Able to maintain neutral pelvis while doing single leg lift but not able to hold neutral pelvis when doing contralateral arm and leg (20-second hold)

• Fair (3) = Able to do single arm lift and maintain neutral pelvis (20-second hold)

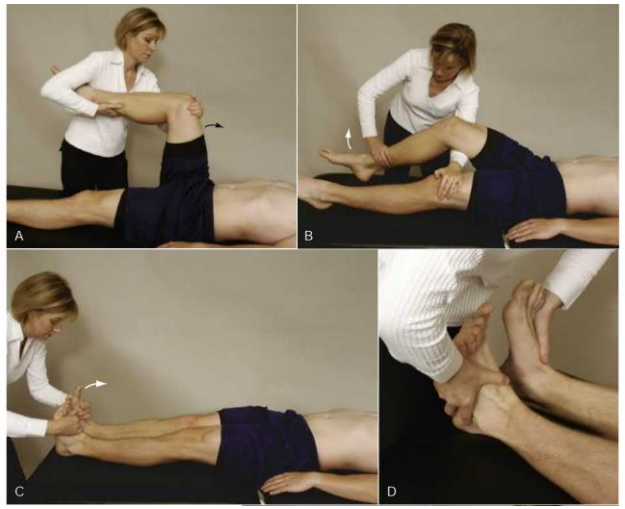
• Poor (2) = Unable to maintain neutral pelvis while doing single arm lift

Neurological Testing

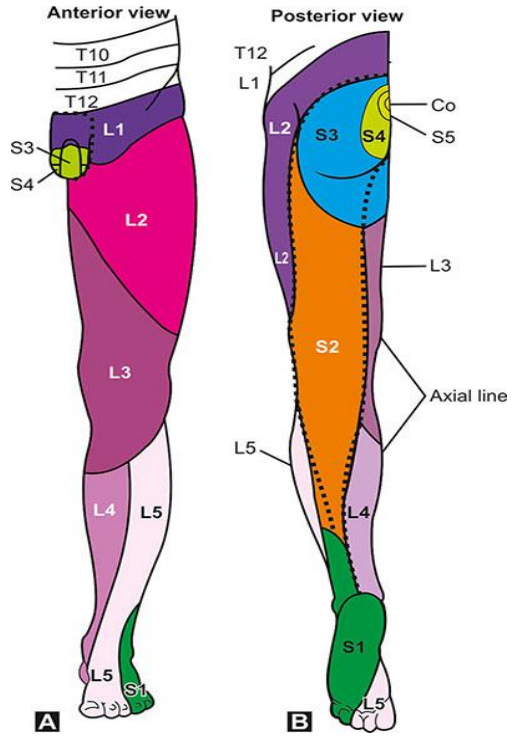
- 1- Dermatomes : L1-S3
- 2- Myotomes: L2-S2
- 3- Reflexes: L3-S1

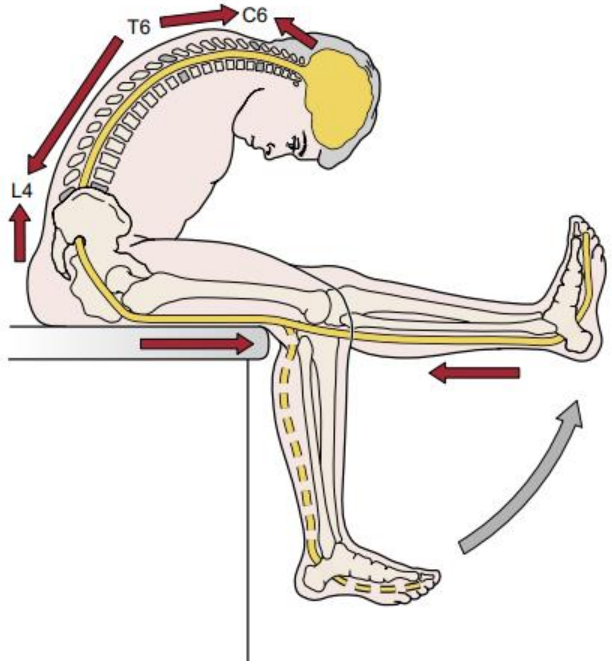
- SLR test, Slump Test and femoral tension test

- Babinski test
- Clonus sign
- Hoffman’s
- Lhermitte's sign

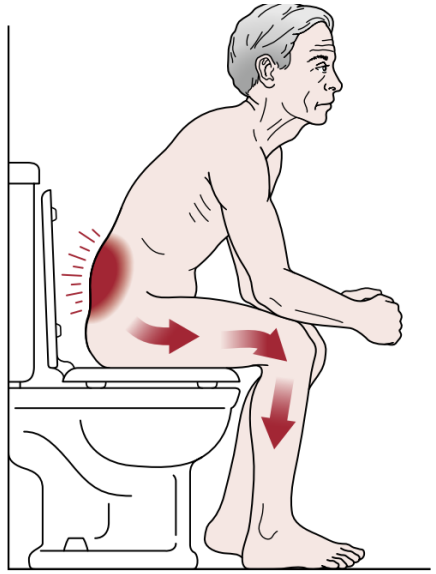


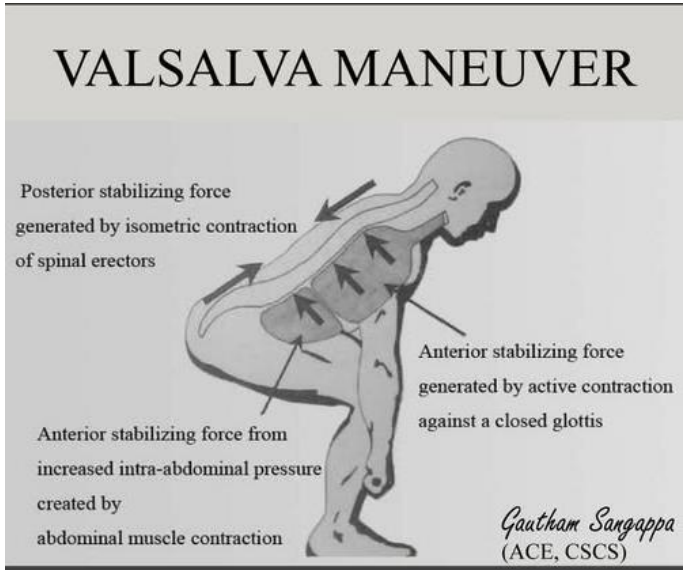






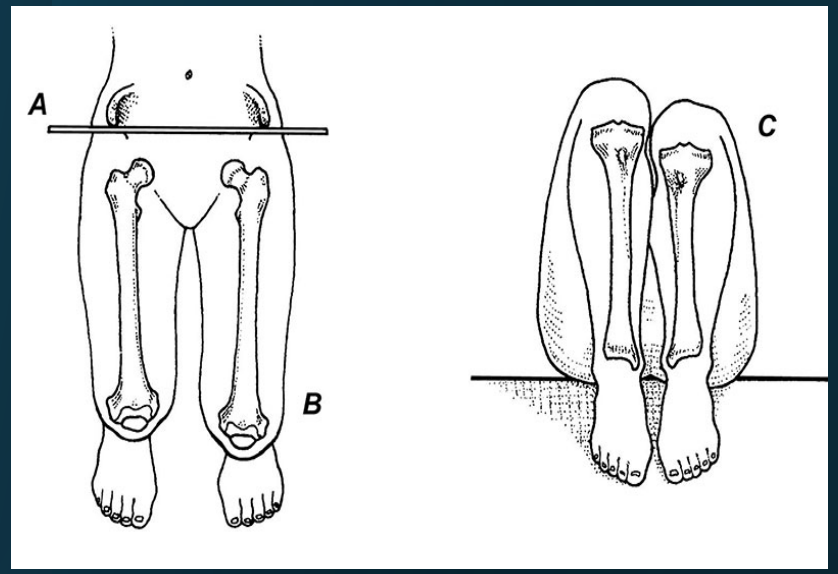
Valsalva Maneuver for radicular pain

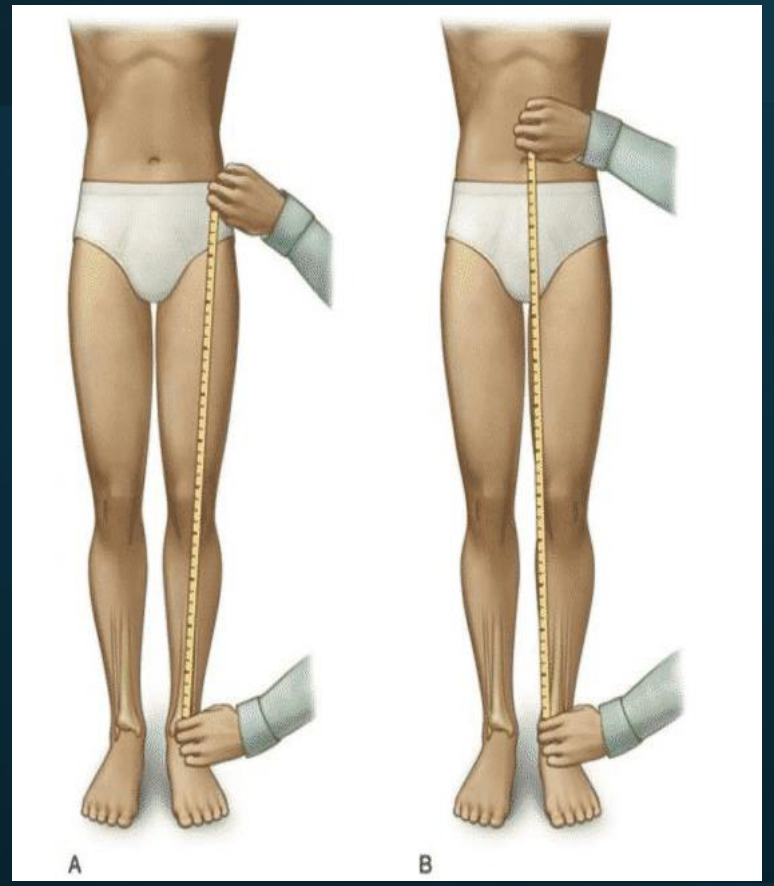


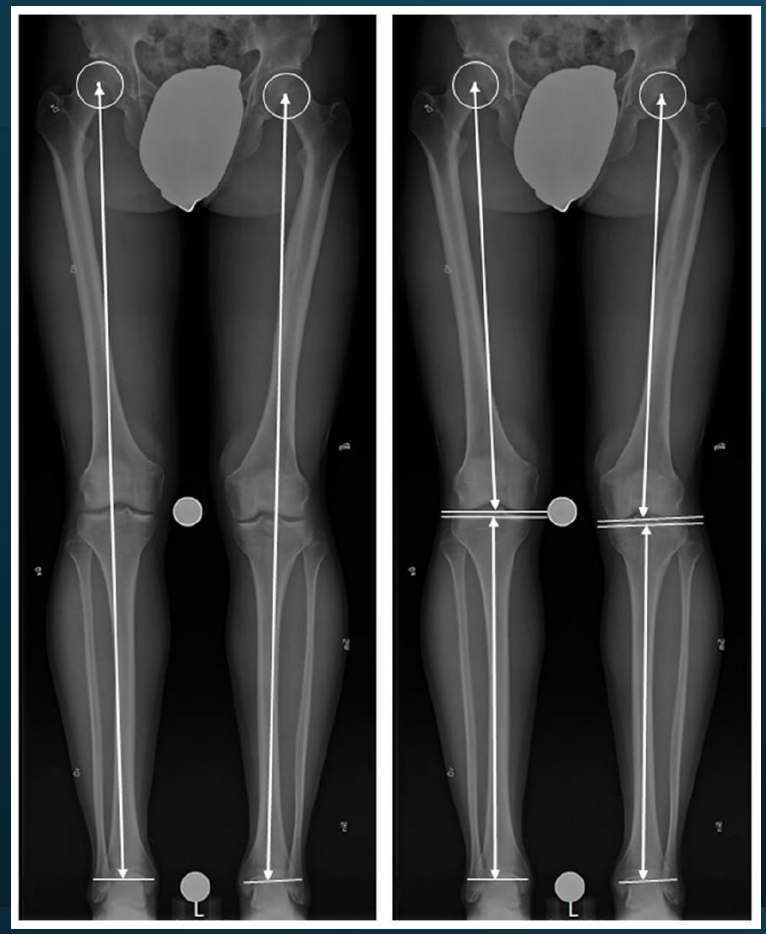




Leg length discrepancy







Special tests

Lumbo Sacral Spine

Special testing

- **Tests for Lumbar Instability**
- (a) Prone instability test,
- (b) Passive lumbar extension test





Facet (Zygapophyseal) Joint Involvement (Clinical Prediction Rule)

(1) localized unilateral back pain;

(2) replication/aggravation of pain by unilateral pressure over facet joint or transverse process;

(3) lack of nerve root symptoms;

(4) if referred, pain does not extend beyond the knee;

(5) pain is eased in flexion;

(6) reduced movement on side of facet joint pain;

(7) unilateral muscle spasm over facet joint; and

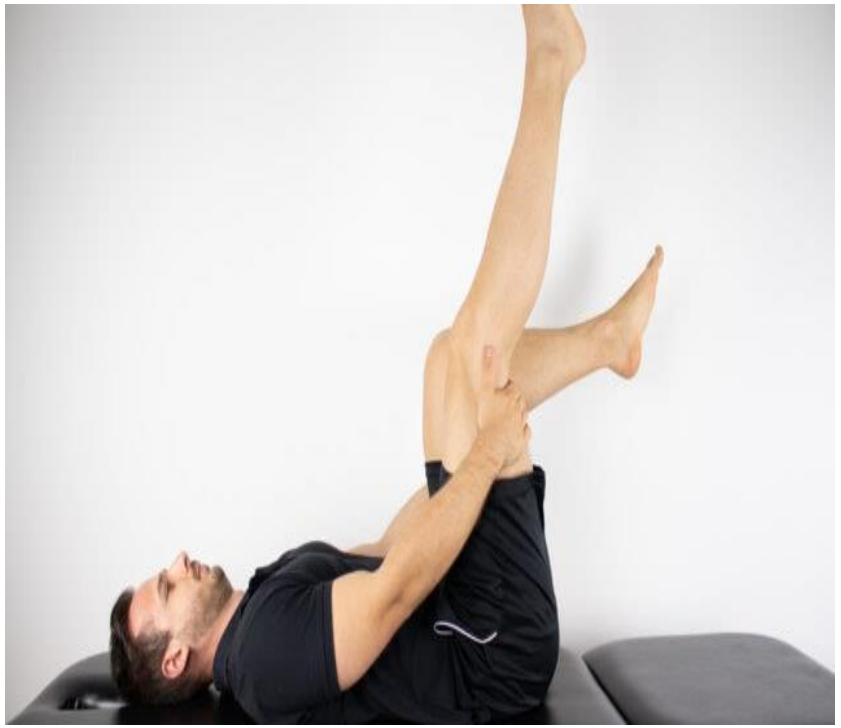
(8) pain on extension and extension with lateral flexion or rotation to same side

lumbar zygapophyseal joint Dysfunction

- Quadrant (Lumbar Quadrant, Extension Quadrant, Kemp’s test)



Tests for Muscle Tightness



90–90 Straight Leg Raising Test for tight hamstrings.

Ober’s Test for tight tensor fasciae latae.

Rectus Femoris Test (Ely’s test).

Modified Thomas Test for tight iliopsoas/ quadriceps.

Laslett Cluster of SIJ Provocation Tests

- Thigh Thrust Test
- Distraction Test
- Compression Test
- Sacral Thrust Test
- Gaenslen's Test

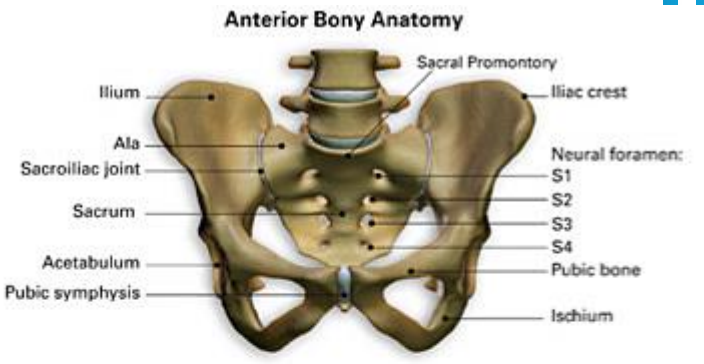
- Diagnostic accuracy of the Laslett cluster:
- Three or more positive tests that reproduce the patient's familiar pain indicates SIJ as the likely source.
- Sensitivity approximately 91%.
- Specificity approximately 78%.
- Gold standard comparison: fluoroscopically guided SIJ injection.

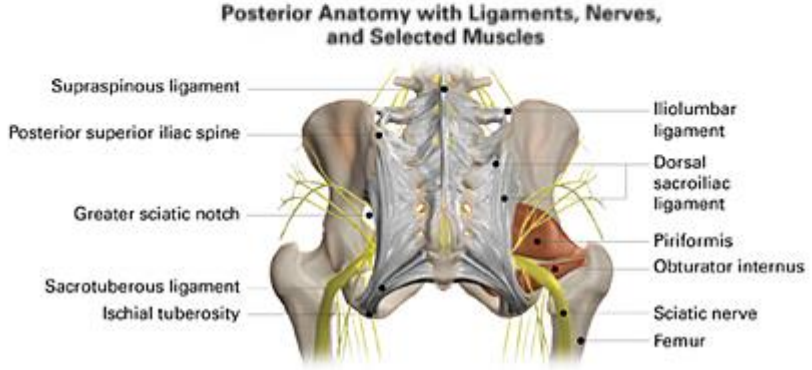
Sacroiliac Joint (SIJ)

- Sacroiliac joint (SIJ) pain is one of the most common causes of low back pain, accounting for 15 to 30% of all cases.
- It is the most likely source of low back pain in patients after having undergone either lumbar or lumbosacral fusion surgery.

Symptoms of SIJ Dysfunction After Lumbar Fusion:

- Pain in the lower back or buttocks (often localized around the sacroiliac joints).
- Reduced range of motion in the lower back, pelvis, or hips.
- Pain with prolonged sitting or standing, especially on one leg.
- Pain when changing positions (e.g., moving from sitting to standing, or from lying down to sitting).





Cluster of Laslett

1- Sacral Thrust test

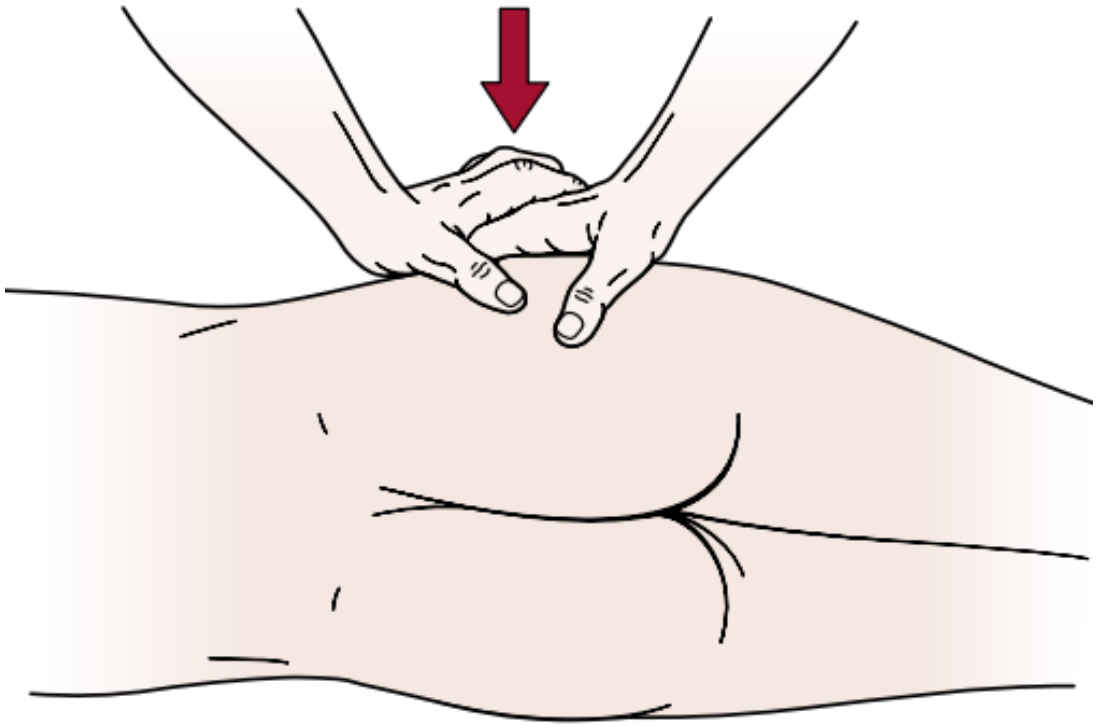


2- Distraction Test





4- Thigh Thrust test



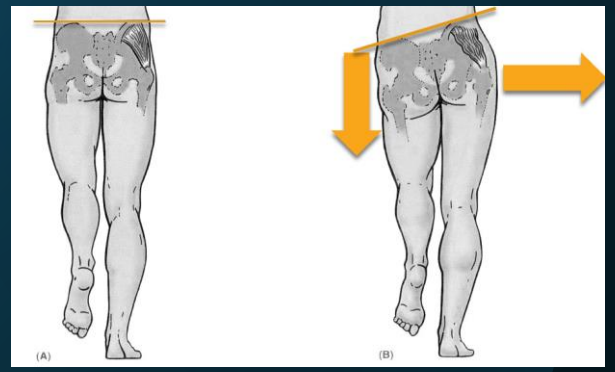
3- Approximation (compression) test

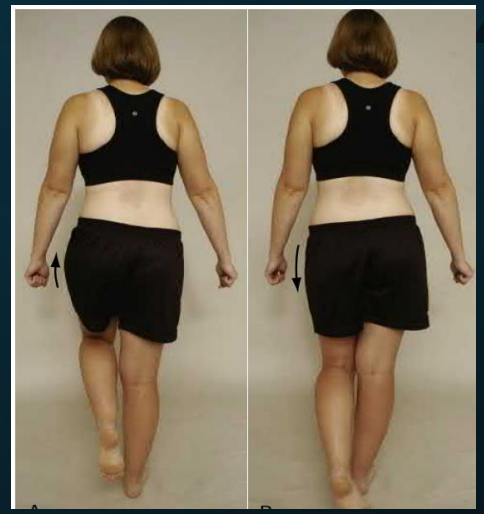
Gaenslen's Test





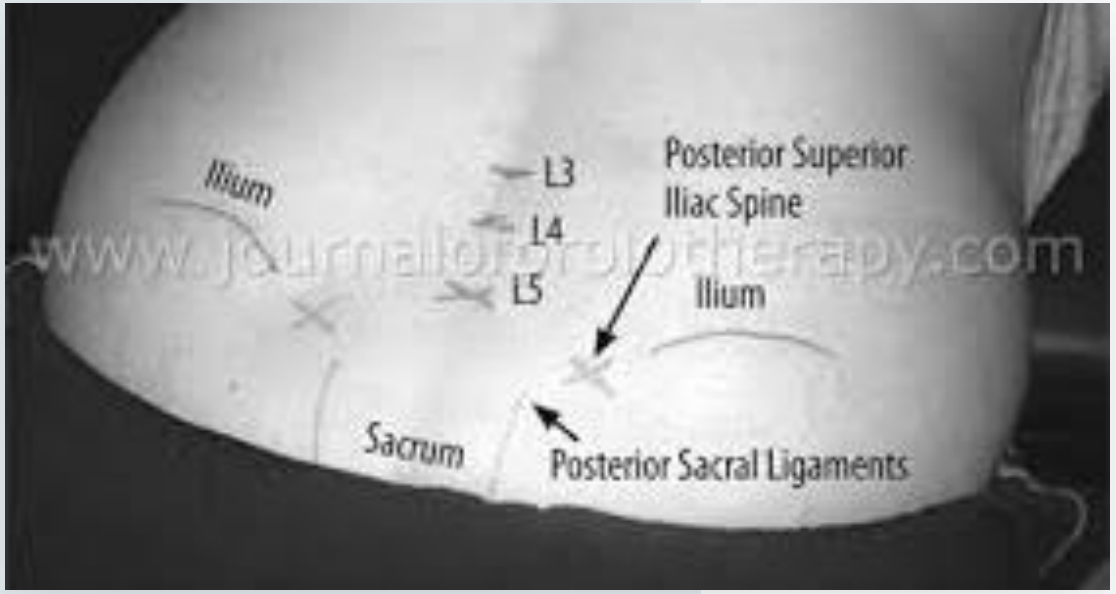
Trendelenburg sign



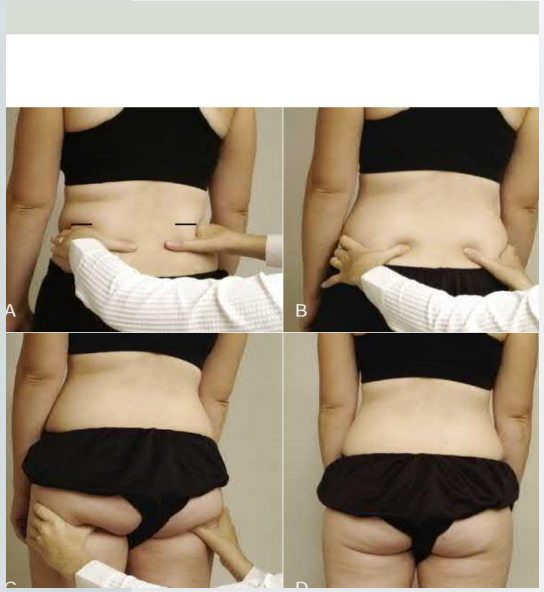


- **Negative Test (Normal):** The pelvis on the non-weight-bearing (unsupported) side remains level or slightly rises. This indicates strong and functional hip abductor muscles on the standing leg side.

- **Positive Test (Abnormal):** The pelvis on the non-weight-bearing side drops or tilts downward. This signifies weakness or dysfunction of the gluteus medius and minimus muscles on the *standing* leg side. The patient may also lean their trunk toward the stance leg to compensate for the imbalance.

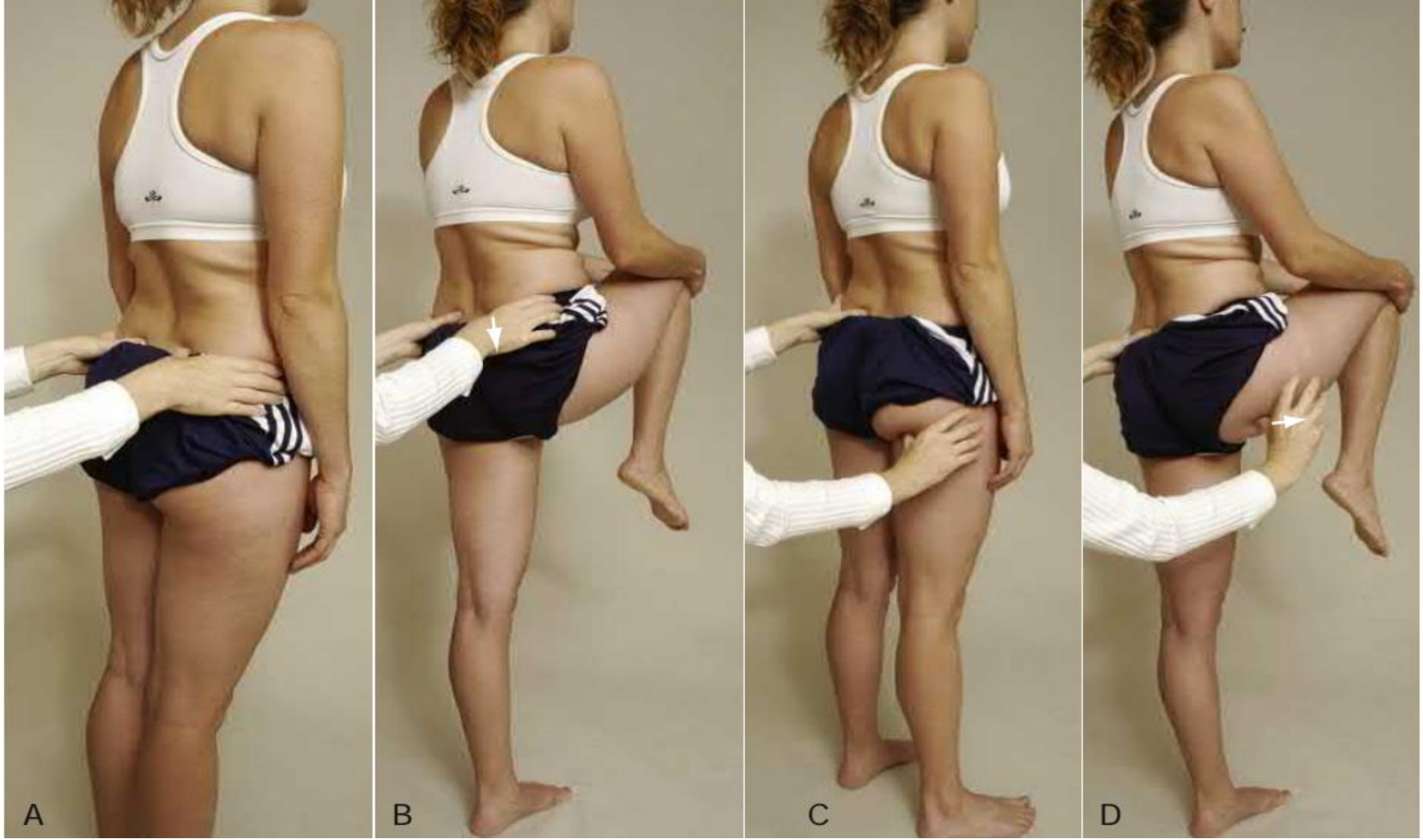


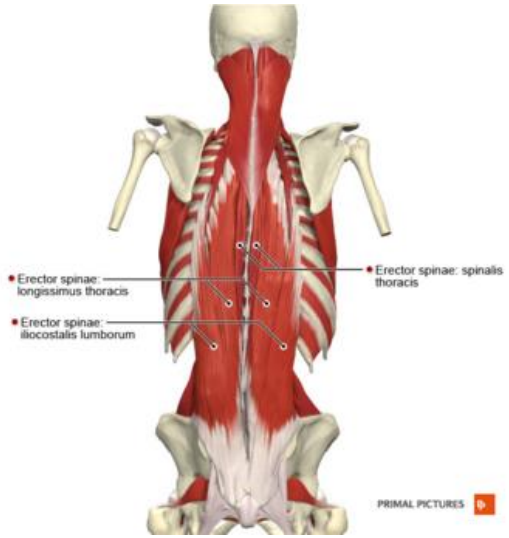




Palpation

1. Spinous processes
2. Transverse processes
3. Last rib
4. ASIS
5. Pubic bone
6. Inguinal area
7. Iliac Crest
8. Sacrum
9. PSIS
10. Gluteal fold
11. Ischial Tuberosities
12. Greater Trochanter
13. Iliac Crest


A
B
C
D









Palpation of Piriformis muscle and the erector spinae group (spinalis, longissimus, and iliocostalis)

Quadratus Lumborum and the glutes





Intermittent Claudication

Neurogenic claudication:

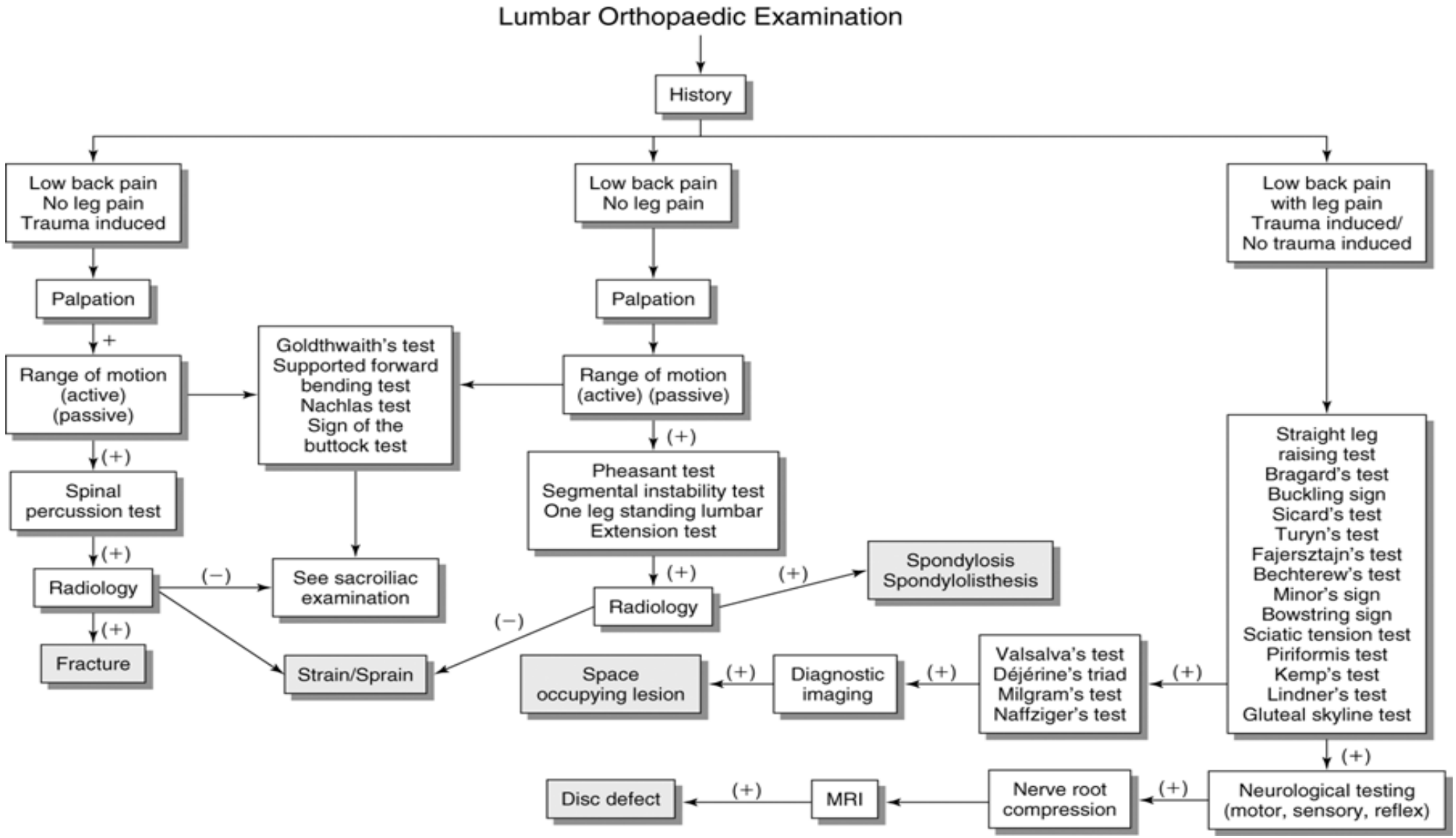
- Bicycle test
- Treadmill Test
- Stoop test

- Intermittent claudication Intermittent claudication is a pain or cramping in the legs that occurs during exercise and is relieved by rest. It implies arterial insufficiency to the tissues. It is most commonly evident when activity occurs because of the increased vascular demand of the tissues.
- There are two types of intermittent claudication—vascular and neurogenic.
- The vascular type is most commonly the result of arteriosclerosis, arterial embolism, or thrombo-angiitis obliterans and commonly manifests itself with symptoms in the legs.
- The neurogenic type is sometimes called pseudoclaudication or cauda equina syndrome and is commonly associated with spinal stenosis and its effect on circulation to the spinal cord and cauda equina.
- The symptoms in this case may be manifested in the back or sciatic nerve distribution.



Functional Assessment

1. Quick Test (the lower peripheral joints)
2. Squatting
3. Single leg stance
4. Forward reach
5. Toe walk (S1)
6. Heels walk (L4-5)


Lumbar Orthopaedic Examination
History
Low back pain
No leg pain
Trauma induced
Palpation
+
Range of motion
(active)
(passive)
(+)
Spinal
percussion test
(+)
Radiology
(−)
See sacroiliac
examination
(+)
Fracture
Strain/Sprain
Goldthwaith's test
Supported forward
bending test
Nachlas test
Sign of the
buttock test
Low back pain
No leg pain
Palpation
Range of motion
(active) (passive)
(+)
Pheasant test
Segmental instability test
One leg standing lumbar
Extension test
(+)
Radiology
(+)
Spondylosis
Spondylolisthesis
(−)
Low back pain
with leg pain
Trauma induced/
No trauma induced
Straight leg
raising test
Bragard's test
Buckling sign
Sicard's test
Turyn's test
Fajersztajn's test
Bechterew's test
Minor's sign
Bowstring sign
Sciatic tension test
Piriformis test
Kemp's test
Lindner's test
Gluteal skyline test
(+)
Valsalva's test
Déjérine's triad
Milgram's test
Naffziger's test
(+)
Diagnostic
imaging
(+)
Space
occupying lesion
(+)
Neurological testing
(motor, sensory, reflex)
(+)
Nerve root
compression
(+)
MRI
(+)
Disc defect